

Robust Bias-Corrected Inference in RDDs

Conditional Robustness and Local-Support Limits in Finite Samples



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Date: 23 June 2026

Main Reference Paper: Calonico, Cattaneo and Titiunik (2014), *Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs*

Contribution Statement: This seminar project has two complementary parts. My project partner studies whether RBC inference can be extended beyond sharp RDDs to fuzzy RD and regression kink designs. This paper keeps the sharp RDD framework fixed and examines when RBC remains reliable in finite samples, especially under heteroskedasticity, non-normal errors, and limited local support.

Main text length: approximately 44,389 characters, including spaces.

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1. Introduction

Regression discontinuity (RD) designs identify treatment effects by comparing local differences in conditional expectations on either side of a cutoff and have become an important tool for causal inference in empirical research. RD estimation is typically based on local polynomial regression, in which bandwidth choice determines the estimator's bias-variance trade-off. In practice, researchers often use an MSE-optimal bandwidth that minimizes the mean squared error of the point estimator, thereby balancing smoothing bias against sampling variance. Although this bandwidth is well suited to point estimation, it does not make smoothing bias negligible relative to the standard error. Conventional confidence intervals constructed directly at this bandwidth therefore often exhibit substantial undercoverage. Calonico, Cattaneo, and Titiunik (2014) proposed robust bias-corrected (RBC) inference, which explicitly estimates and corrects smoothing bias and uses robust Studentization to incorporate uncertainty from bias estimation into the standard error, thereby delivering asymptotically valid inference with an MSE-optimal bandwidth.

RBC has become a central method for continuity-based RD inference, but the range over which its finite-sample advantages remain valid is still an important practical question. Specifically, does its coverage improvement remain stable under heteroskedastic, skewed, or heavy-tailed errors? When the running variable is discrete, can RBC still estimate and correct local bias reliably? More importantly, when RBC performance begins to deteriorate, is the issue a problem of confidence-interval construction, or an estimation problem caused by insufficient information near the cutoff?

To answer these questions, this paper uses the DGP constructed by CCT (2014) from the data of Ludwig and Miller (2007) as the Baseline and implements a unified paired Monte Carlo design. The Baseline first compares Conventional MSE, undersmoothing, bias-corrected, RBC-MSE, and RBC-CER confidence intervals to reproduce the core mechanisms underlying RBC. It then introduces three extensions: heteroskedasticity, non-normal errors, and a discrete running variable. Each extension changes only one central dimension of the Baseline and, wherever possible, uses the same underlying random samples, so that changes across scenarios

can be attributed to a clearly defined failure mechanism.

The simulation results show that the robustness of RBC first appears in its ability to accommodate changes in the sampling environment. In the Baseline, RBC-MSE raises coverage from 85.24% for Conventional MSE to 93.00%, while avoiding the substantial interval expansion associated with undersmoothing. Across alternative heteroskedasticity structures, RBC coverage remains between 92.2% and 93.3%; under skewed, heavy-tailed, and contaminated errors, it remains between 93.0% and 95.9%. These results indicate that, as long as local support near the cutoff remains adequate, neither heteroskedasticity nor error non-normality by itself eliminates the coverage advantage of RBC.

A discrete running variable reveals a more fundamental limitation. When the discrete grid becomes denser as the sample size increases, RBC continues to deliver stable coverage. Under fixed coarse support, however, a larger sample merely places more observations on existing mass points without increasing the number of distinct support points. Local support may therefore decline as the bandwidth shrinks. A further failure-frontier analysis traces how RBC moves through a support-warning region, coverage deterioration, and operational failure. Although RBC generally deteriorates later than conventional inference, bias correction itself loses a reliable foundation once local support is too sparse to estimate the regression function and its curvature stably.

The main contribution of this paper is to distinguish, within a unified simulation framework, two fundamentally different forms of pressure on RBC. Heteroskedasticity and non-normal errors primarily create inferential pressure within the local-polynomial framework, whereas fixed coarse support can undermine the informational basis on which that framework rests. The finite-sample performance of RBC can therefore be characterized as conditionally robust: RBC can effectively address smoothing-bias-driven undercoverage, but it cannot substitute for adequate local support or repair an RD design that no longer provides a reliable basis for estimation.

The remainder of the paper is organized as follows. Section 2 explains the smoothing-bias problem in conventional inference, the construction of RBC, and the coverage-error perspective. Section 3 presents the Baseline and the three extension designs. Section 4 reports the Monte Carlo results. Section 5 discusses the scope of RBC, and Section 6 concludes.

2. Robust Bias-Corrected Inference in RD Designs

2.1 Motivation: Traditional Inference Failure

Regression discontinuity design (RDD) is one of the most important quasi-experimental methods in modern causal inference, with its basic idea dating back to Thistlethwaite and Campbell (1960). When treatment status is determined by whether a running variable crosses a known cutoff, units close to the cutoff have very similar values of the running variable and should therefore also be highly comparable in other potential characteristics. Under appropriate continuity conditions, a discontinuous jump in the outcome at the cutoff can be interpreted as a treatment effect.

Hahn, Todd, and van der Klaauw (2001) subsequently established a rigorous nonparametric identification framework for RD designs. In a sharp RD design, for example, treatment status is determined by whether the running variable crosses the cutoff, that is, $D_i = \mathbf{1}\{X_i \geq c\}$. If the conditional expectation functions of the potential outcomes are continuous near the cutoff, the average treatment effect at the cutoff is identified by the difference between the right- and left-hand limits of the conditional expectation of the observed outcome:

$$\tau = \lim_{x \downarrow c} E[Y_i | X_i = x] - \lim_{x \uparrow c} E[Y_i | X_i = x].$$

In empirical applications, this discontinuity is typically estimated using local polynomial regression. For the commonly used local linear estimator, researchers fit separate weighted local linear regressions on the two sides of the cutoff and take the difference between the two intercepts as the RD estimator:

$$\hat{\tau}(h) = \hat{\mu}_+(c) - \hat{\mu}_-(c),$$

where h denotes the bandwidth. Conventional inference based on this estimator typically uses the Wald statistic:

$$T_{\text{conv}} = \frac{\hat{\tau}(h) - \tau}{\text{SE}_{\text{conv}}(h)}.$$

For conventional inference to be valid, the standardized estimator must be centered at the true parameter and approximately follow a standard normal distribution. In local polynomial estimation, however, estimation error reflects not only random sampling variation but also smoothing bias induced by local approximation. More specifically, the standardized statistic can be decomposed as

$$T = \underbrace{\frac{\hat{\tau} - E[\hat{\tau}]}{SE}}_{\rightarrow N(0,1)} + \underbrace{\frac{\overbrace{E[\hat{\tau}] - \tau}^{\text{Smoothing Bias}}}{SE}}_{\text{Scaled Bias}}.$$

Here, the first term represents random sampling error, which can be approximated by a normal distribution under standard regularity conditions; the second represents the remaining effect of smoothing bias measured in standard-error units, commonly referred to as scaled bias. For a local linear RD estimator, the leading smoothing bias in the numerator is of order

$$\text{Bias}[\hat{\tau}(h)] = O(h^2),$$

whereas the standard error in the denominator is of order

$$\text{SE}[\hat{\tau}(h)] = O((nh)^{-1/2}).$$

Consequently, the scaled bias is of order

$$\frac{\text{Bias}[\hat{\tau}(h)]}{\text{se}[\hat{\tau}(h)]} = O(\sqrt{nh} h^2) = O(\sqrt{nh^5}).$$

For conventional Wald inference to be valid, this term must vanish asymptotically, that is,

$$nh^5 \rightarrow 0.$$

This condition is fundamentally a requirement on the bandwidth shrinkage rate: the bandwidth must decrease sufficiently quickly for smoothing bias to be negligible relative to the standard error. In practice, however, bandwidths are generally not selected to satisfy this inferential condition. Researchers more commonly use a data-driven MSE-optimal bandwidth because it targets point-estimation performance by balancing bias and variance.

$$MSE(h) = Bias^2 + Variance \asymp h^4 + \frac{1}{nh}$$

Minimizing this objective function yields an MSE-optimal bandwidth satisfying

$$h_{MSE} \asymp n^{-1/5}.$$

Substituting this bandwidth rate into the condition required for conventional inference, however, gives

$$nh_{MSE}^5 \asymp 1.$$

Thus, under an MSE-optimal bandwidth, smoothing bias decreases as the sample size grows but does not vanish when measured relative to the standard error. The estimator itself can therefore remain consistent, while the conventional Wald statistic is no longer centered correctly and the resulting confidence interval exhibits undercoverage because smoothing bias has not been adequately addressed. In other words, the MSE-optimal bandwidth is natural for point estimation but not for conventional inference.

Before Calonico, Cattaneo, and Titiunik (2014), the principal response to this problem was undersmoothing. Rather than estimating and correcting the bias directly, undersmoothing selects a bandwidth smaller than the MSE-optimal bandwidth so that

$$nh^5 \rightarrow 0.$$

Under this condition, smoothing bias becomes negligible relative to the standard error, allowing conventional Wald inference to regain validity. The cost of undersmoothing, however, is substantial. Artificially reducing the bandwidth lowers the effective number of observations used near the cutoff, raises estimation variance, and generally widens the confidence interval. More importantly, unlike the MSE-optimal bandwidth, undersmoothing does not correspond to a clearly defined optimization criterion and therefore lacks a unified and stable data-driven selection rule in practice.

RD inference before CCT (2014) therefore faced a fundamental tension. An MSE-optimal bandwidth provides a favorable point-estimation trade-off between bias and variance but

produces undercoverage for conventional confidence intervals. Undersmoothing can restore the validity of conventional inference, but only at the cost of higher variance and wider intervals. This tension provided the direct motivation for robust bias-corrected inference.

2.2 The CCT Solution: Bias Correction and Robust Studentization

Robust bias-corrected (RBC) inference, proposed by Calonico, Cattaneo, and Titiunik (2014), resolves this tension. Its central idea is straightforward: if coverage failure arises from smoothing bias that remains under the MSE-optimal bandwidth, there is no need to eliminate that bias by shrinking the bandwidth; it can instead be estimated and corrected directly. This approach preserves the lower-variance advantage of the MSE-optimal bandwidth while restoring valid inference. More specifically, RBC inference consists of two related steps:

Step 1: Recentering the Statistic

CCT first characterize the bias. Let $\hat{\tau}_1(h)$ denote the local linear RD estimator obtained using the main bandwidth h . Its leading smoothing bias can be written as

$$\text{Bias}(\hat{\tau}_1(h)) = h^2 B + o(h^2),$$

where B is determined by the second derivatives of the regression functions on the two sides of the cutoff. Because B is not directly observable, CCT estimate it using a local quadratic regression. This requires a second bandwidth, the pilot bandwidth b . Unlike h , b is not used to re-estimate the treatment effect; its purpose is to estimate the curvature term required for bias correction. After obtaining an estimate of the leading bias using b , one can construct the bias-corrected estimator:

$$\hat{\tau}_{bc}(h, b) = \hat{\tau}_1(h) - h^2 \hat{B}(h, b).$$

This step recenters a statistic that would otherwise be displaced from the true value by smoothing bias and can therefore be understood as recentering the statistic.

Step 2: Robust Studentization

Simply subtracting the estimated bias, however, is not enough. $\hat{B}(h, b)$ is itself an estimated random variable and therefore introduces additional uncertainty into the final estimator. If the conventional standard error is retained after bias correction, the confidence interval may still be too narrow. CCT therefore construct a robust variance estimator that incorporates both the sampling variability of the bias estimate and its covariance with the original estimator:

$$\hat{V}_{rbc}(h, b) = \widehat{Var}[\hat{\tau}_1(h) - h^2 \hat{B}(h, b)].$$

This yields the RBC t-statistic:

$$T_{rbc} = \frac{\hat{\tau}_{bc}(h, b) - \tau}{\widehat{se}_{rbc}(h, b)}.$$

The corresponding RBC confidence interval is

$$CI_{rbc} = \left[\hat{\tau}_{bc}(h, b) \pm z_{1-\frac{\alpha}{2}} \widehat{se}_{rbc}(h, b) \right].$$

The key to RBC inference is therefore not bias subtraction alone, but the simultaneous correction of both center and scale. Bias correction adjusts the center of the statistic, while robust Studentization adjusts its scale. The former addresses the non-negligible smoothing bias under an MSE-optimal bandwidth, and the latter accounts for the additional randomness introduced by bias estimation.

2.3 Why Is RBC Preferable? A Coverage-Error Perspective

The preceding section presented the construction of RBC inference and explained how it resolves the inferential difficulty associated with an MSE-optimal bandwidth. What remains is a structural explanation for its superior performance. This section therefore adopts the coverage-error perspective developed by Calonico, Cattaneo, and Farrell (2018), shifting attention from the construction of the method to the higher-order sources of confidence-interval coverage error.

Intuitively, the central limit theorem establishes only that, as the sample size tends to infinity, the distribution of the Studentized statistic can be approximated by the standard normal distribution:

$$P(T_n \leq z) = \Phi(z) + o(1).$$

In finite samples, however, an approximation error remains between the true distribution and the standard normal distribution. An Edgeworth expansion can represent its structure as

$$P(T_n \leq z) = \Phi(z) + n^{-1/2}A(z) + n^{-1}B(z) + \dots .$$

For the simplest sample-mean problem, the higher-order terms in an Edgeworth expansion are typically related to distributional features such as skewness and kurtosis. These terms characterize the sources of finite-sample normal-approximation error and explain why the actual coverage of a confidence interval may differ from its nominal level. In local polynomial RD inference, however, the approximation error has a more complex structure. Following the coverage-error expansion in CCF (2018), this paper conceptually groups the principal higher-order sources of finite-sample coverage error into three channels: distributional approximation, scaled residual bias, and Studentization.

This decomposition clarifies the source of the advantage of RBC over undersmoothing. Undersmoothing mainly reduces scaled bias by shrinking the bandwidth until smoothing bias becomes asymptotically negligible. RBC, by contrast, directly corrects the leading smoothing bias and uses robust Studentization to incorporate the additional uncertainty introduced by bias estimation into the standard error. RBC therefore acts not only on scaled bias but on several components of coverage error related to both bias and Studentization.

The coverage-error perspective also provides a directional framework for the subsequent Monte Carlo analysis: when the main sources of coverage error are subjected to more severe finite-sample stress, does RBC continue to improve confidence-interval coverage relative to conventional inference?

Finally, the coverage-error perspective motivates an extension in bandwidth selection. The RBC procedure of CCT (2014) primarily addresses the fact that smoothing bias is non-negligible under an MSE-optimal bandwidth. Yet the MSE-optimal bandwidth itself is designed

to minimize the mean squared error of a point estimator rather than to optimize the coverage performance of a confidence interval. Calonico, Cattaneo, and Farrell (2020) therefore use the coverage-error expansion to show that an RBC confidence interval can instead employ a bandwidth targeted more directly at coverage error: the coverage-error-rate-optimal, or CER-optimal, bandwidth. The central distinction between RBC-CER and RBC-MSE thus lies not in the inference procedure itself but in the bandwidth objective. Their finite-sample performance is compared in the Baseline simulation below.

3. Monte Carlo Design

3.1 Baseline Design

This paper takes as its central reference the DGP constructed in CCT (2014) from the data of Ludwig and Miller (2007). Specifically, the running variable is generated as

$$X_i = 2B_i - 1, B_i \sim \text{Beta}(2,4)$$

which implies $X_i \in [-1,1]$, and the cutoff is set at $c = 0$. The outcome satisfies

$$Y_i = \mu_2(X_i) + \varepsilon_i,$$

where

$$\varepsilon_i \sim N(0, 0.1295^2).$$

The underlying regression function is defined as

$$\mu_2(x) = \begin{cases} 3.71 + 2.30x + 3.28x^2 + 1.45x^3 + 0.23x^4 + 0.03x^5, & x < 0, \\ 0.26 + 18.49x - 54.81x^2 + 74.30x^3 - 45.02x^4 + 9.83x^5, & x \geq 0. \end{cases}$$

The true treatment effect at the cutoff is therefore

$$\tau = \mu_2(0^+) - \mu_2(0^-) = 0.26 - 3.71 = -3.45.$$

This DGP is well suited to serve as the Baseline for two reasons. First, it contains a clear treatment-effect discontinuity at the cutoff. Second, the regression function exhibits substantial

local curvature near the cutoff. Consequently, conventional inference with an MSE-optimal bandwidth is readily affected by smoothing bias, making the role of RBC inference especially transparent. The design is therefore not an idealized, perfectly smooth setting, but a practically challenging benchmark for the stress designs that follow.

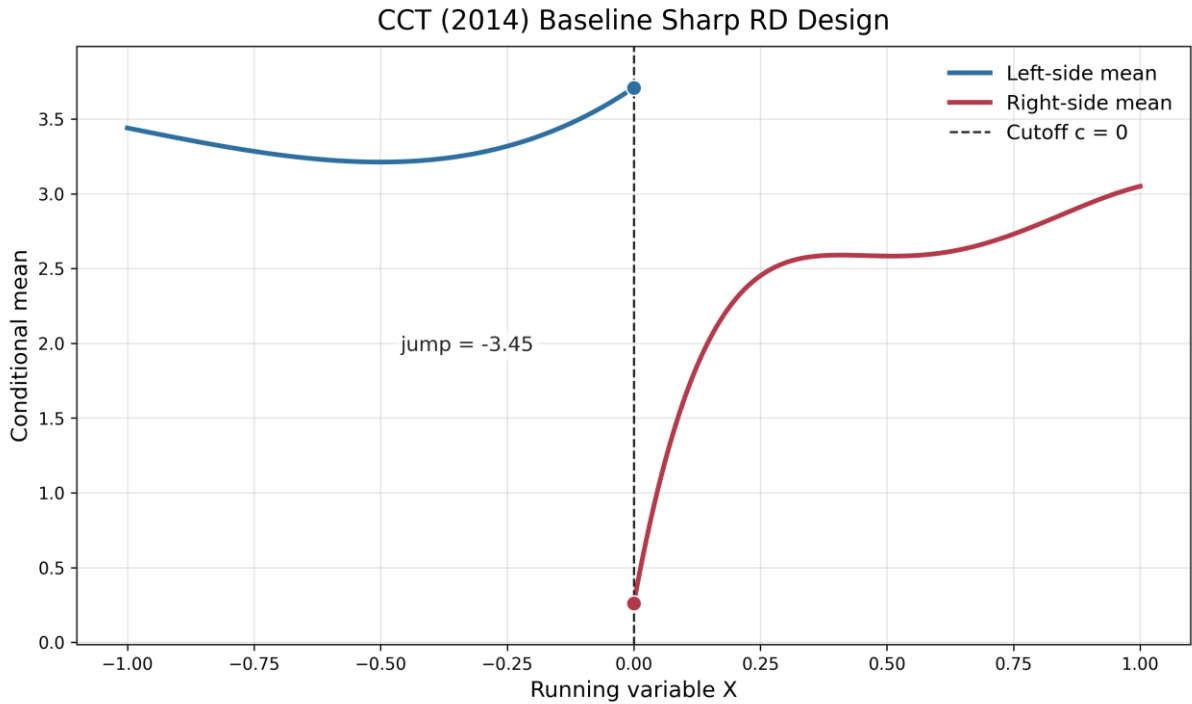


Figure 3.1. Baseline regression function in the CCT sharp RD design

3.2 Extension Designs

The coverage-error perspective in Section 2.3 provides the starting point for the Monte Carlo analysis. When the standard conditions required for local polynomial estimation hold, the finite-sample performance of RBC can be understood in terms of sampling error, scaled bias, and Studentization error. From this perspective, the heteroskedasticity and non-normality extensions examine whether changes in the conditional variance and the error distribution weaken the ability of RBC to control coverage error.

The discrete-running-variable extension pushes the analysis further by asking whether the local estimation conditions underlying coverage-error analysis continue to hold as the number

of distinct support points near the cutoff declines. The simulations thus move from testing the robustness of RBC within the local-polynomial framework to identifying the boundary of that framework itself.

All extensions are built on the Baseline DGP. To preserve comparability across simulation results, each extension changes only one central dimension of the DGP. This design makes it possible to link changes in RBC performance to a specific mechanism, rather than to the combined effect of several simultaneous changes. Table 3.1 summarizes the structure of the simulation design, the Baseline assumption relaxed by each extension, and the corresponding primary comparison.

Table 3.1. Simulation Map: Baseline, Stress Dimensions, and Relevant Comparisons

Scenario	Assumption Relaxed	RBC Component	Comparison
Baseline	None	Benchmark	Conventional, Undersmoothing, Bias-corrected, RBC- MSE, and RBC-CER
Heteroskedasticity	Homoskedasticity	Variance estimation and Studentization	Conventional vs RBC
Non-normality	Gaussian errors	Finite-sample Distribution approximation	Conventional vs RBC
Discrete (X)	Continuous support	Local support and bias estimation	Conventional vs RBC vs bias-aware curvature-bound interval

The main methodological comparison is between the 95% Conventional MSE CI and the 95% RBC MSE CI, allowing the analysis to assess whether the coverage gain of RBC over

conventional inference persists across the different stress designs. Unless otherwise stated for the discrete-running-variable extension, all simulations use a sample size of $n=500$ and $R=5,000$ Monte Carlo repetitions. The paper also adopts a paired simulation design: all methods or subscenarios to be compared are computed from the same underlying random draws, thereby reducing the influence of simulation noise on the comparisons.

3.2.1 Heteroskedasticity Design

From a coverage-error perspective, heteroskedasticity can affect confidence-interval coverage primarily through standard-error estimation and Studentization. To examine this mechanism, this section changes only the Baseline error term from homoskedastic to conditionally heteroskedastic:

$$\varepsilon_i = 0.1295\sqrt{w_s(X_i)}u_i, \quad u_i \sim N(0,1),$$

where $w_s(X_i)$ is the normalized variance multiplier, defined as

$$w_s(X_i) = \frac{v_s(X_i)}{n^{-1} \sum_{j=1}^n v_s(X_j)}.$$

Here, $v_s(x)$ denotes the raw variance structure. Because the sample mean of $w_s(X_i)$ is normalized to one, the average conditional variance of the error term in each simulation remains equal to 0.1295 squared. Thus, the design does not alter the overall scale of the noise; it changes only the shape of the error variance along the running variable.

Four variance structures are considered:

$$H_0: v_0(x) = 1,$$

$$H_1: v_1(x) = 1 + x^2,$$

$$H_2: v_2(x) = 1 + \mathbf{1}\{x \geq 0\},$$

$$H_3: v_3(x) = 1 + \exp(-|x|/0.10),$$

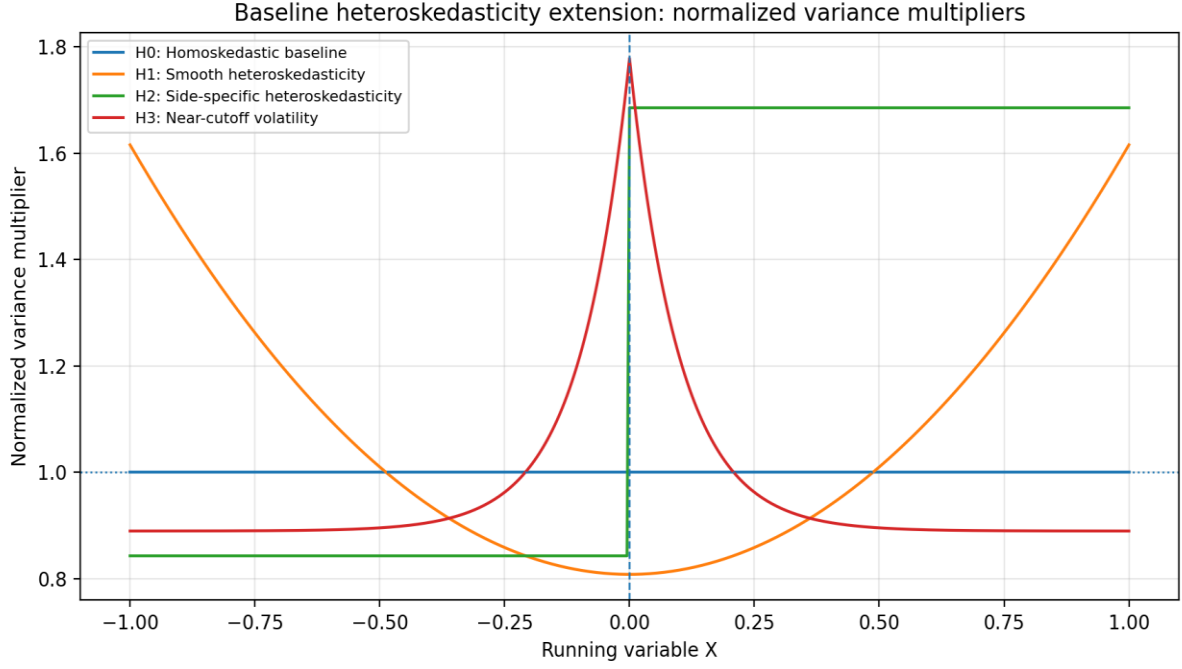


Figure 3.2. Normalized variance multipliers in the heteroskedasticity extension

Here, H_0 is the homoskedastic Baseline; H_1 allows the variance to rise as the running variable moves away from the cutoff; H_2 imposes different error variances on the two sides of the cutoff; and H_3 raises the variance near the cutoff. Both the Conventional CI and the RBC CI reported in this paper use heteroskedasticity-robust standard errors. The purpose of this section is therefore not to test whether unadjusted standard errors fail under heteroskedasticity, but to examine whether changes in the conditional variance structure still weaken the finite-sample coverage of RBC once heteroskedasticity-robust variance estimation is already employed.

3.2.2 Non-Normal Error Design

From a coverage-error perspective, non-normal errors can affect confidence-interval coverage primarily by changing the finite-sample distributional approximation of the Studentized statistic. The second extension therefore considers non-normal error terms. The outcome is specified as

$$Y_i = \mu_2(X_i) + \varepsilon_i, \quad \varepsilon_i = 0.1295U_i,$$

and all scenarios satisfy

$$E(U_i) = 0, \text{Var}(U_i) = 1.$$

Differences across the non-normal designs therefore arise from the shape of the error distribution rather than from changes in the scale of the error term. This construction keeps the non-normal scenarios comparable with the Baseline in terms of error variance.

Four error-distribution scenarios are considered:

$$N0: U_i = Z_i, Z_i \sim N(0,1),$$

$$N1: U_i = \frac{\chi_{1,i}^2 - 1}{\sqrt{2}},$$

$$N2: U_i = \frac{T_{3,i}}{\sqrt{3}}, T_{3,i} \sim t_3,$$

$$N3: \tilde{U}_i = Z_i + \delta D_i, D_i \sim \text{Bernoulli}(\pi),$$

$$U_i = \frac{\tilde{U}_i - \pi\delta}{\sqrt{1 + \pi(1 - \pi)\delta^2}}, \pi = 0.02, \delta = 10.$$

Original error distributions by scenario

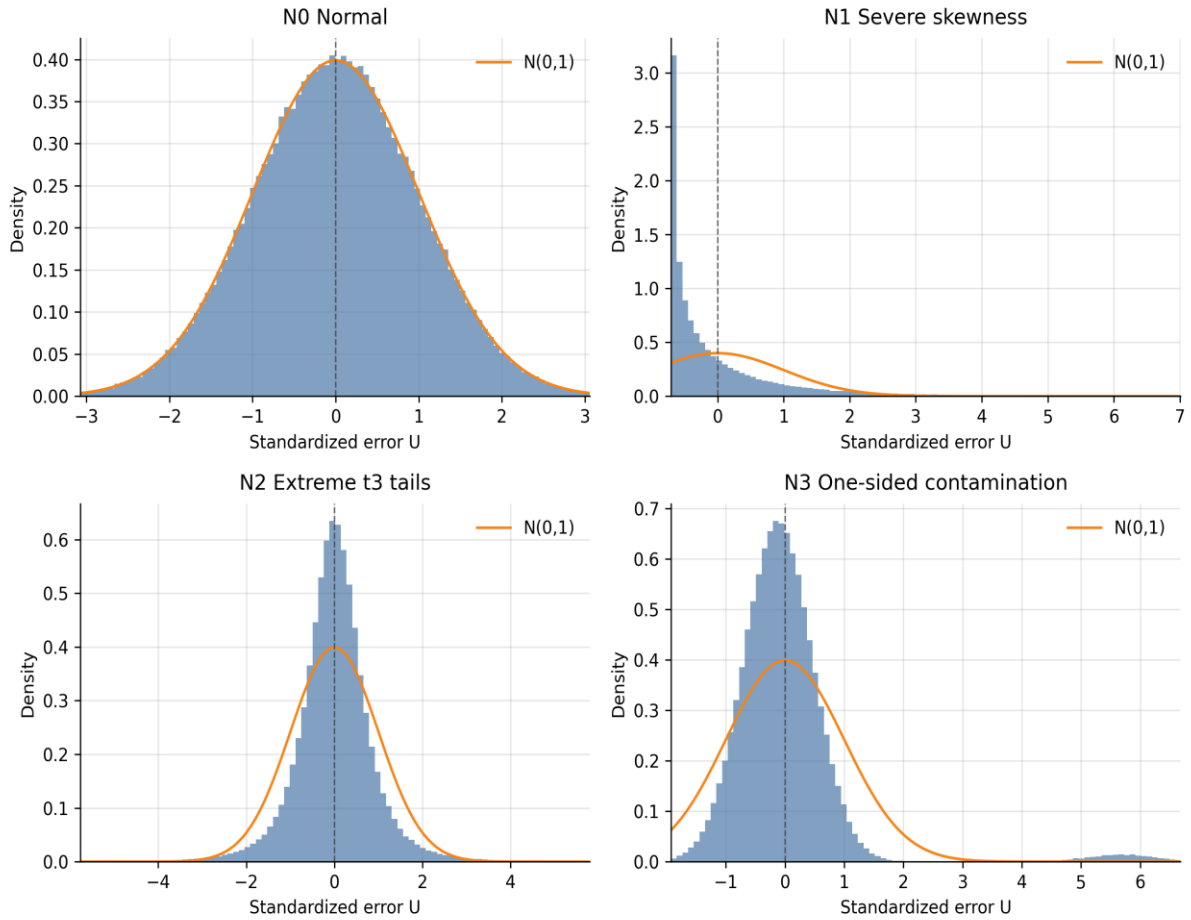


Figure 3.3. Error distributions in the non-normal extension

The error-distribution plots show that these designs are not mild departures from normality but severe non-normal stress tests. $N0$ is the normal-error Baseline; $N1$ uses a standardized χ_1^2 distribution to introduce right-skewed errors; $N2$ uses a standardized t_3 distribution to examine heavy-tailed errors; and $N3$ is a one-sided contamination design in which approximately 2% of observations receive large positive shocks, thereby mimicking rare one-sided outliers.

3.2.3 Discrete Running Variable Design

The third extension examines the effect of discretizing the running variable on RBC inference. Unlike the preceding two designs, discretization directly changes the local support structure near the cutoff. When there are too few distinct support points, not only confidence-interval inference but also local linear estimation and RBC bias correction may lose their conventional interpretation. To distinguish finite-sample performance under ordinary discretization from the specific failure boundary under coarse discrete support, this section uses two connected experiments.

Both experiments build on the Baseline DGP and change only the observed form of the running variable. A continuous latent running variable X_i is first generated and then mapped to a midpoint grid with spacing δ :

$$X_i^{obs}(\delta) = \delta \left(\left\lfloor \frac{X_i}{\delta} \right\rfloor + \frac{1}{2} \right), \quad \delta > 0.$$

When $\delta = 0$ ($X_i^{obs}(0) = X_i$), the design coincides with the continuous Baseline. The outcome is generated as

$$Y_i = \mu_2(X_i^{obs}) + \varepsilon_i$$

where

$$\varepsilon_i \sim N(0, 0.1295^2),$$

The simulation target remains the latent discontinuity of $\mu_2(x)$ at zero

$$\tau = -3.45.$$

The first experiment studies sample-size paths under general discretization. It begins with a fixed-support design in which the grid spacing remains unchanged as the sample size increases:

$$\delta \in \{0.01, 0.015, 0.02, 0.025, 0.03, 0.035, 0.04, 0.045, 0.05\}.$$

The corresponding shrinking-support design allows the grid spacing to decrease as the sample size increases:

$$\delta_n = 0.05 \sqrt{\frac{500}{n}}.$$

The sample sizes are

$$n \in \{500, 750, 1000, 1500, 2000, 3000, 4000, 5000\}.$$

This experiment distinguishes two cases. Along the fixed-support path, a larger sample only increases the number of observations at existing mass points. Along the shrinking-support path, a larger sample is accompanied by denser local support. The experiment therefore tests whether increasing the sample size alone is sufficient to relax the local-support constraint created by a discrete running variable.

The second experiment then conducts a discrete failure-frontier analysis to trace how RBC moves from normal operation into regions of insufficient local support and eventual estimation failure as the grid spacing δ increases. Unlike the first experiment, which compares fixed- and shrinking-support paths across sample sizes, the second experiment fixes $n = 500$ and evaluates the continuous benchmark $\delta = 0$, a refined fixed-support grid from $\delta = 0.01$ to 0.08 in increments of 0.005 , and the additional coarse-grid values $\delta \in \{0.09, 0.10, 0.12\}$.

To characterize the local support actually available to RBC, the local support score for repetition r is defined as

$$S_r = \min \{M_{h,L,r}, M_{h,R,r}, M_{b,L,r}, M_{b,R,r}\},$$

Here, $M_{h,L,r}$ and $M_{h,R,r}$ denote the numbers of distinct support points to the left and right of the cutoff within the main bandwidth h , while $M_{b,L,r}$ and $M_{b,R,r}$ denote the corresponding counts within the bias-correction bandwidth b . Because effective implementation of RBC

depends on all four components of local support, S_r takes their minimum, thereby recording the weakest support component in each repetition and helping identify regions of adequate support, insufficient support, and estimation failure.

For the RBC interval, a repetition is classified as minimally usable when the estimator returns a finite, correctly ordered confidence interval with a positive length below the prespecified pathology cutoff of 10, when each side contains at least two distinct support points within the main bandwidth h , and when each side contains at least three distinct support points within the bias-correction bandwidth b . The preferred-support diagnostic is deliberately stricter: it requires at least 20 observations on each side within h , at least five distinct support points on each side within h , and at least six distinct support points on each side within b . Scenario-level coverage is calculated among minimally usable repetitions and is interpreted as an ordinary Monte Carlo coverage rate only when the minimal-usability rate is at least 80% and at least 30 minimally usable repetitions remain.

As a supplementary comparison, the discrete extension also reports a bias-aware curvature-bound interval inspired by Kolesár and Rothe (2018). Instead of estimating and subtracting bias as in RBC, this interval enlarges the conventional local-linear confidence interval by an explicit worst-case bias bound. In the present Monte Carlo design, the curvature bound is constructed from the known simulation DGP, so the interval should be interpreted as a bias-aware sensitivity benchmark rather than as a fully data-driven empirical procedure.

4. Monte Carlo Results

4.1 Baseline Results

This section first examines whether the core mechanism of CCT (2014) can be reproduced under the present design. Table 4.1 reports the Monte Carlo results for five confidence intervals under the Baseline design.

Table 4.1. Baseline Monte Carlo results

Method	Coverage (MCSE)	Avg. interval length	Bias	RMSE	Bandwidth choice
Conventional MSE	85.24% (0.0050)	0.315	0.050	0.098	mserd (0.1)
Undersmoothing	91.50% (0.0039)	0.485	0.018	0.120	$0.5 \times \text{mserd}$ (0.05)
Bias-corrected MSE	90.48% (0.0042)	0.315	0.011	0.086	mserd (0.1)
RBC-MSE	93.00% (0.0036)	0.344	0.011	0.086	mserd (0.1)
RBC-CER	92.74% (0.0037)	0.396	0.009	0.097	cerrd (h=0.073)

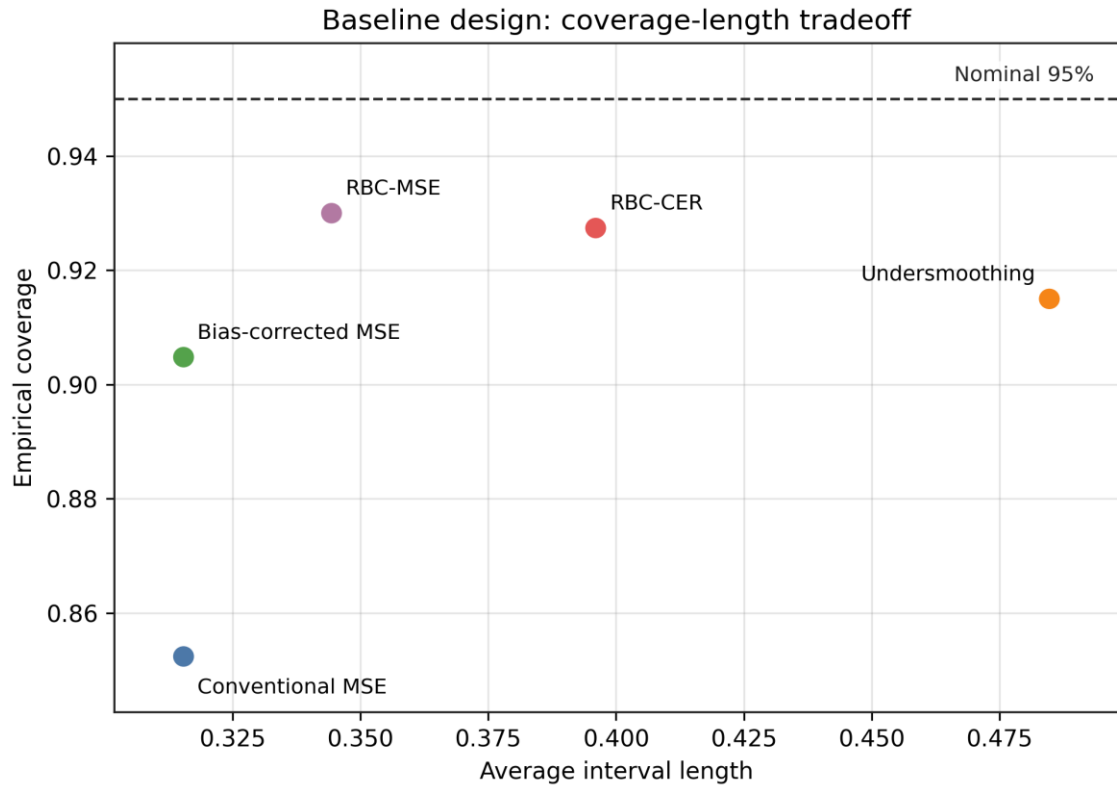


Figure 4.1. Coverage-length tradeoff under the baseline design

The results first show substantial undercoverage for the Conventional MSE confidence interval. Its empirical coverage is only 85.24%, approximately 9.76 percentage points below the nominal 95% level. This result is consistent with the theoretical discussion above: although the MSE-optimal bandwidth is appropriate for the bias-variance trade-off in point estimation, it does not make smoothing bias negligible relative to the standard error under conventional inference. The center of the interval is therefore systematically displaced, producing pronounced undercoverage. The coverage bars in Figure 4.2 likewise show that Conventional MSE has the largest gap from the nominal 95% coverage rate.

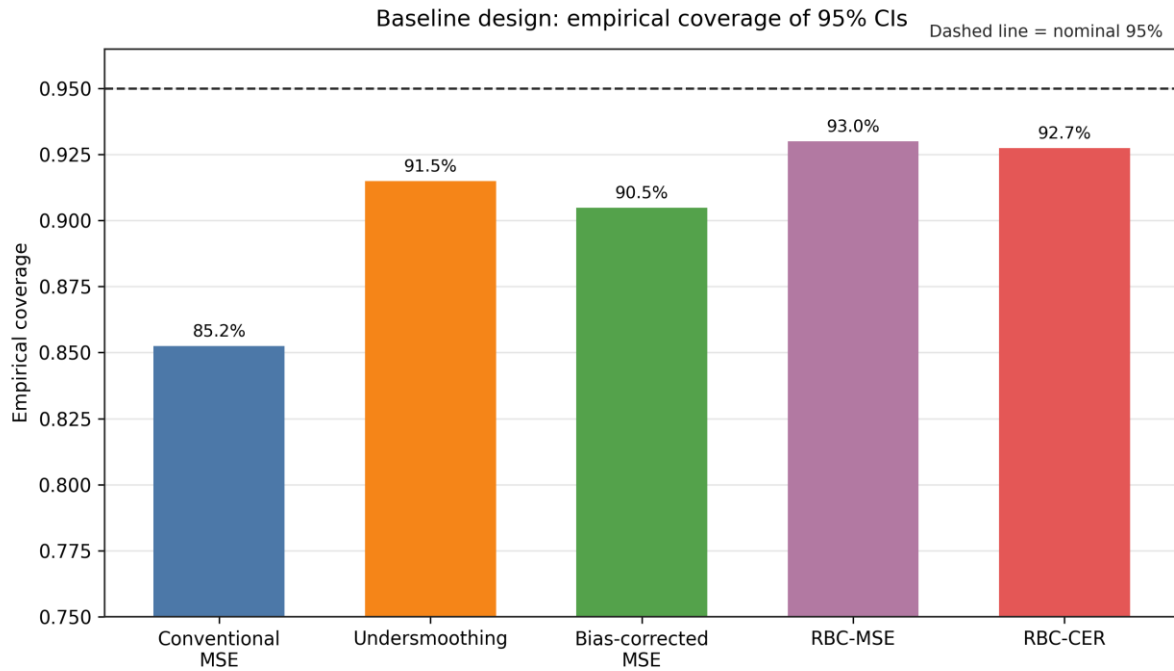


Figure 4.2. Empirical coverage of 95% confidence intervals under the baseline design

Undersmoothing raises coverage to 91.50%, confirming that a smaller bandwidth can attenuate the effect of smoothing bias. The cost of this improvement is substantial, however. Because the undersmoothing bandwidth used here is one-half of the original MSE-optimal bandwidth, the average bandwidth falls from approximately 0.100 to 0.050, reducing the effective local sample and increasing the average interval length to 0.485, the longest among all methods. Its RMSE is also 0.120, higher than that of the other methods. The coverage improvement from undersmoothing therefore comes primarily from sacrificing estimation precision and widening the interval rather than from more efficient finite-sample estimation.

The Bias-corrected MSE results further show that bias correction alone does not solve the inferential problem. The bias-corrected estimator reduces finite-sample bias from 0.050 under Conventional MSE to 0.011, indicating that the estimated leading-bias adjustment is effective. Its coverage, however, is only 90.48%, slightly below that of undersmoothing. Correcting only the center of the interval is therefore insufficient: bias estimation introduces additional uncertainty, and the confidence interval may remain too narrow unless the standard error is adjusted accordingly.

RBC-MSE achieves the highest observed coverage and the most favorable coverage–length trade-off in this simulation. It uses the same bias-corrected center as Bias-corrected MSE, so both have a bias of 0.011, but RBC additionally employs robust Studentization to incorporate uncertainty from bias estimation into the standard error. As a result, RBC coverage rises to 93.00%, higher than for any other method. Its average interval length is 0.344, substantially shorter than the 0.485 under undersmoothing. The coverage-length trade-off in Figure 4.1 likewise places RBC in a more favorable position: it delivers the largest coverage improvement at a comparatively small cost in interval length.

RBC-MSE coverage is nevertheless 93.00%, still below the nominal 95% level. Decomposing noncoverage by direction shows that, of the total 7.00% noncoverage rate, 4.98% occurs because the entire confidence interval lies above the true treatment effect, whereas only 2.02% occurs because the interval lies below it. The remaining coverage error is therefore concentrated in an upward displacement of the interval. This asymmetry is consistent with directional higher-order residual bias generated by the complex local shape of the Baseline DGP near the cutoff.

Finally, relative to RBC-MSE, RBC-CER selects a smaller main bandwidth: the average value of h falls from 0.100 to 0.073, while the average interval length increases from 0.344 to 0.396, reflecting a more conservative bandwidth choice oriented toward coverage error. Its empirical coverage is 92.74%, however, and therefore does not exceed the 93.00% achieved by RBC-MSE. In the present finite sample and DGP, simply changing the bandwidth objective from MSE to coverage error is not sufficient to eliminate the remaining coverage gap; local curvature and other higher-order finite-sample errors may also contribute to that gap.

Overall, the Baseline simulation reproduces the central mechanism of CCT (2014). Conventional MSE inference exhibits substantial undercoverage under an MSE-optimal bandwidth. Undersmoothing improves coverage, but at the cost of longer confidence intervals and greater estimation uncertainty. The advantage of RBC arises from the combined effect of bias correction and robust Studentization. The subsequent extensions use Conventional MSE and RBC-MSE from the Baseline as reference methods to examine the scope of RBC under heteroskedasticity, non-normal errors, and a discrete running variable.

4.2 Heteroskedasticity

Table 4.2. Monte Carlo results under heteroskedasticity

Variance structure	Conventional coverage	RBC coverage	Conventional IL	RBC IL
H0: Homoskedastic baseline	85.2%	93.0%	0.315	0.344
H1: Smooth heteroskedasticity	85.5%	93.3%	0.296	0.322
H2: Side-specific heteroskedasticity	84.5%	92.5%	0.339	0.372
H3: Near-cutoff volatility	85.0%	92.2%	0.375	0.412

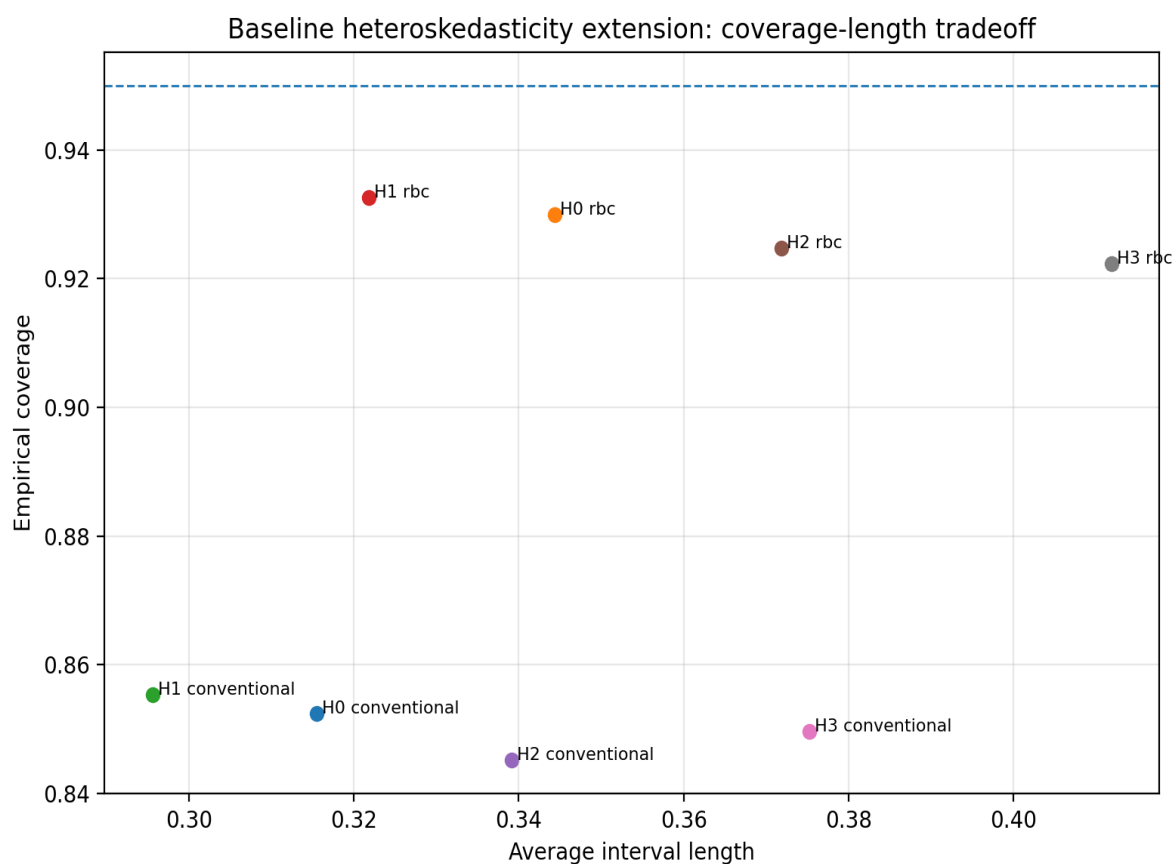


Figure 4.3. Coverage-length tradeoff under heteroskedasticity

The Conventional MSE confidence interval exhibits substantial undercoverage under all four variance structures. Even with heteroskedasticity-robust standard errors, coverage remains only about 84.5%-85.5%, well below the nominal 95% level. The main problem for conventional inference therefore remains non-negligible smoothing bias rather than heteroskedasticity itself.

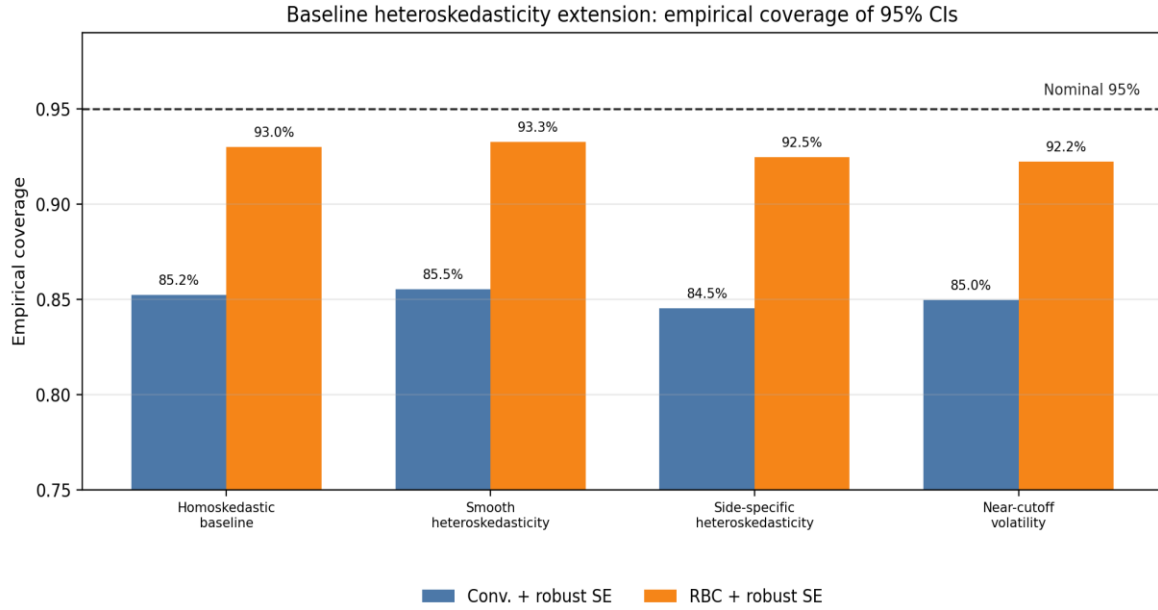


Figure 4.4. Empirical coverage under heteroskedasticity

By contrast, the RBC confidence interval substantially improves coverage under every variance structure. Across H0-H3, RBC coverage remains between 92.2% and 93.3%. More importantly, this improvement is stable across all three heteroskedastic alternatives. In the present designs, heteroskedasticity itself therefore does not constitute an additional systematic source of failure for RBC inference.

In terms of interval length, RBC intervals are consistently longer than Conventional MSE intervals. This is expected because robust Studentization accounts not only for uncertainty in the original estimator but also for the additional uncertainty introduced by bias correction. Differences in interval length across variance structures also reflect the local nature of RD estimation. H1 produces the shortest average intervals because normalization leaves the error

variance relatively low near the cutoff. H3 produces the longest intervals because the error variance rises near the cutoff, precisely where RD estimation places the greatest weight.

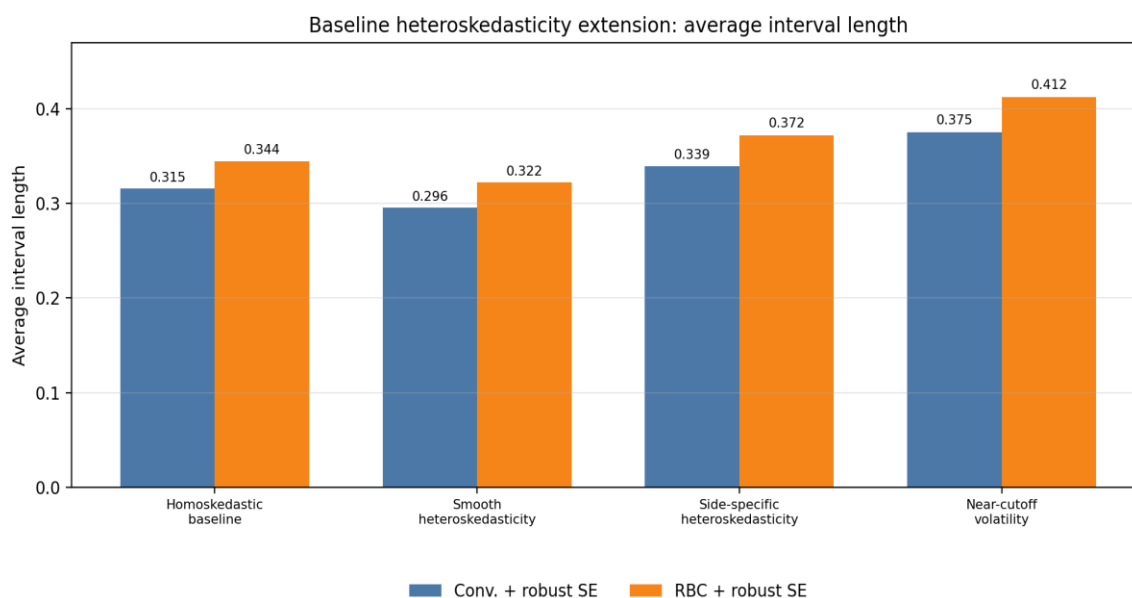


Figure 4.5. Average interval length under heteroskedasticity

Overall, the heteroskedasticity extension shows that, when robust standard errors are used, changes in the conditional variance structure do not materially weaken the coverage performance of RBC inference.

4.3 Non-Normal Errors

This section next examines whether RBC inference depends on normality of the error term.

Table 4.3 reports the main results for the non-normal extension.

Table 4.3. Monte Carlo results under non-normal errors

Scenario	Conventional coverage	RBC coverage	Conventional IL	RBC IL
N0: Normal	85.2%	93.0%	0.315	0.344
N1: Severe skewness	86.1%	94.7%	0.303	0.330
N2: Extreme tails	87.3%	94.7%	0.299	0.326
N3: One-sided contamination	87.4%	95.9%	0.293	0.319

Non-normal extension: coverage-length tradeoff

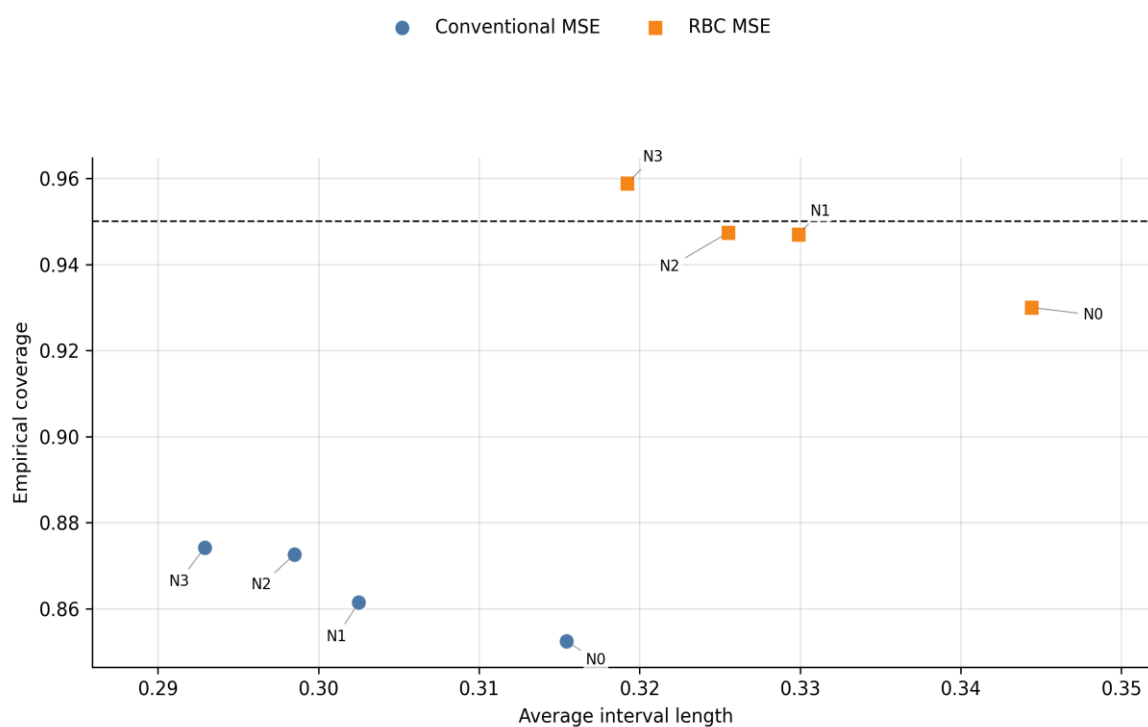


Figure 4.6. Coverage-length tradeoff under non-normal errors

The Conventional MSE confidence interval continues to exhibit substantial undercoverage in every non-normal scenario. Across N0-N3, its empirical coverage is only 85.24%-87.42%, well below the nominal 95% level.

By contrast, the RBC confidence interval substantially improves coverage in every scenario. RBC coverage rises from 93.00% under N0 to 94.70% under N1 and 94.74% under N2, reaching 95.88% under N3. Figure 4.7 shows that RBC coverage is clearly higher than Conventional MSE coverage under every error distribution and is consistently closer to the nominal 95% level. In the present design, the finite-sample performance of RBC inference therefore does not depend primarily on normality of the error term itself. More precisely, RBC requires the distribution of the bias-corrected, Studentized statistic to be well approximated by a normal distribution, not the raw errors themselves to be normally distributed. The empirical distributions of the RBC t-statistic in Figure 4.8 are consistent with this interpretation.

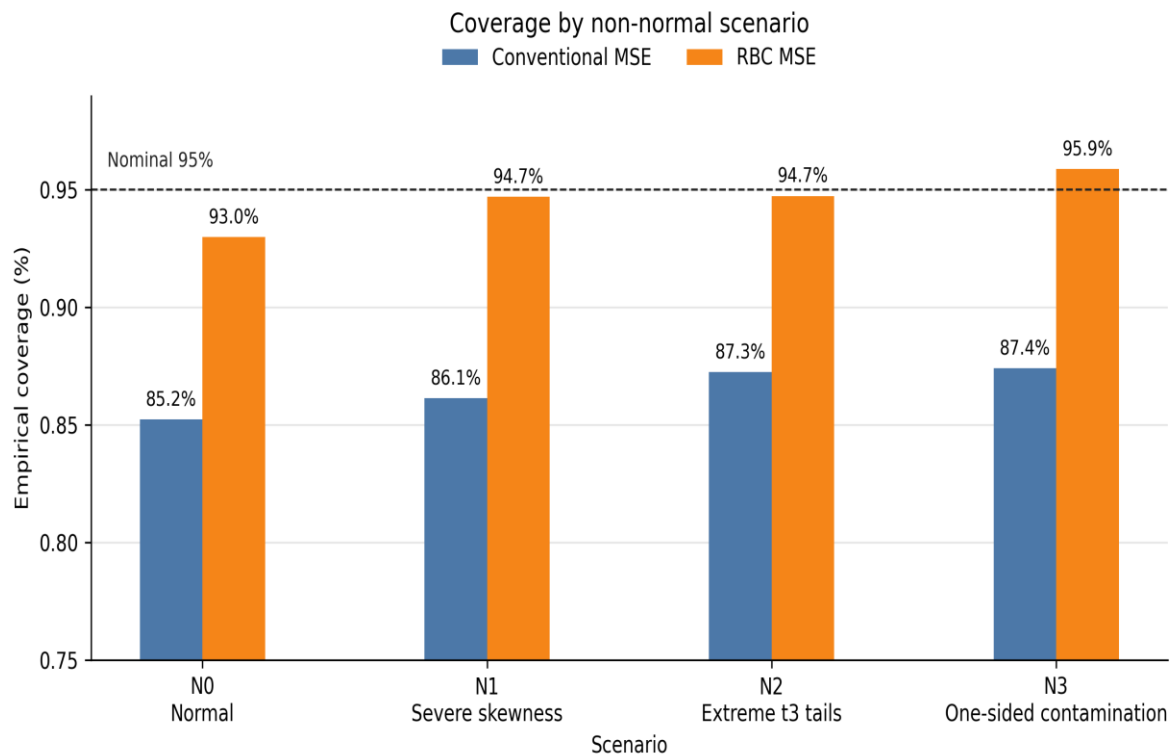


Figure 4.7. Coverage by non-normal scenario

RBC t-statistic distributions by scenario

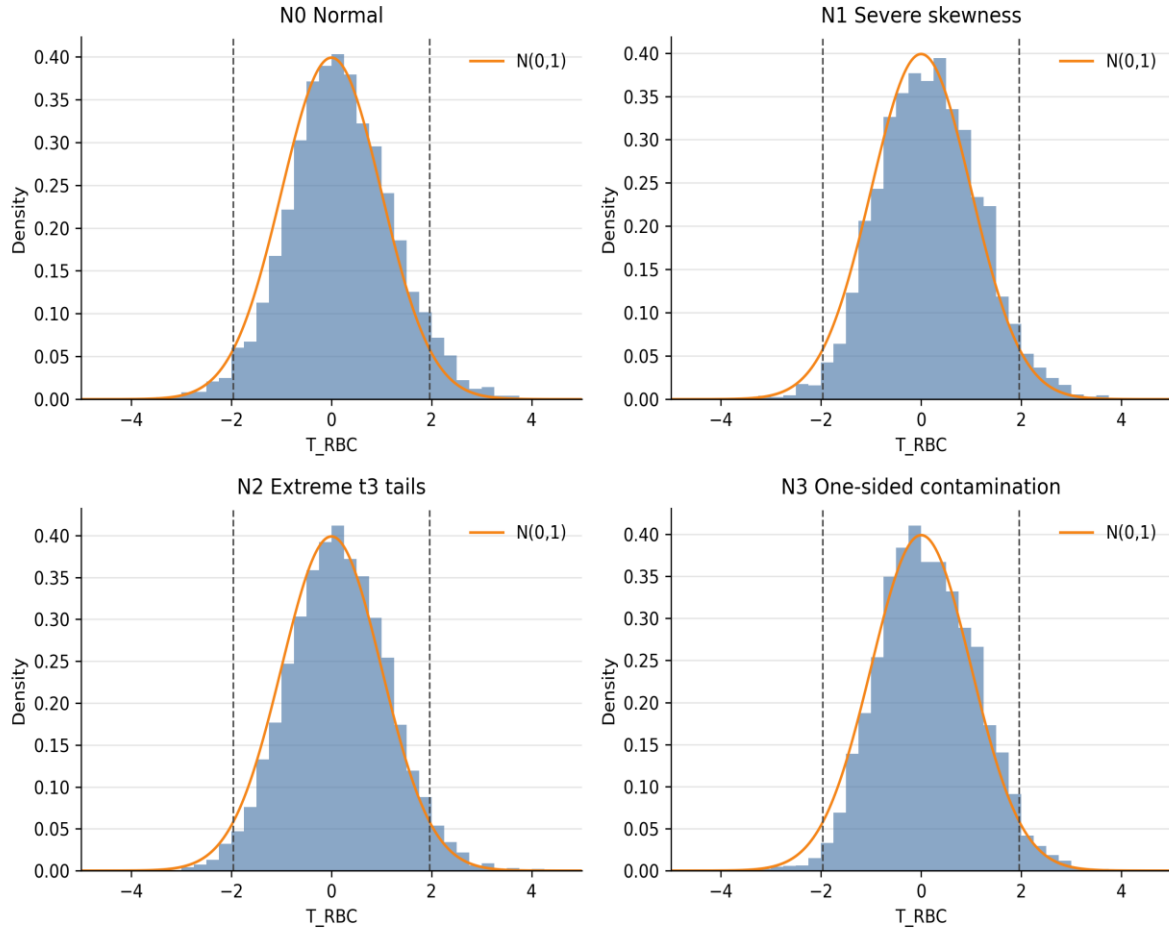


Figure 4.8. RBC t-statistic distributions under non-normal errors

One result requires particular explanation: RBC coverage does not decline under the severe non-normal scenarios and is instead higher than under the normal Baseline. This should not be interpreted as a general claim that non-normal errors improve RBC inference. It must be understood in light of the direction of noncoverage in the Baseline. The remaining RBC coverage error is concentrated in intervals lying above the true treatment effect, and Figure 4.9 shows that this component falls from 4.98% under N0 to 4.02% under N1, 3.56% under N2, and 3.14% under N3. The increase in coverage under N1-N3 is therefore better understood as a consequence of the non-normal disturbances altering the finite-sample tail behavior of the Studentized statistic in a way that happens, for this DGP, to reduce the dominant direction of Baseline noncoverage. It is not a general conclusion about non-normal errors.

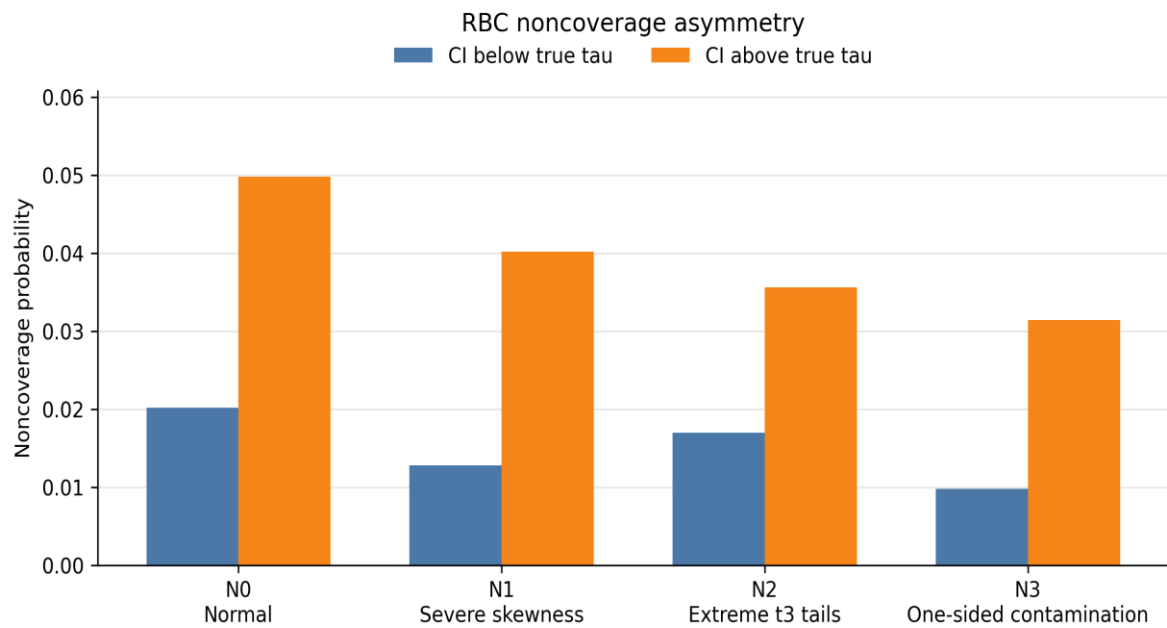


Figure 4.9. RBC noncoverage asymmetry under non-normal errors

Overall, error non-normality is not a principal threat to RBC inference in the designs considered here. RBC continues to correct the undercoverage of Conventional MSE inference in a stable manner.

4.4 Discrete Running Variable

The analysis of the discrete-running-variable results proceeds in two stages. First, the general discretization experiment compares shrinking-support and fixed-support scenarios to examine RBC coverage and operability across sample sizes. The frontier extension then holds the sample size fixed and considers progressively coarser grid spacings, distinguishing an early conservative warning of inadequate local support from coverage deterioration and complete failure of local estimation.

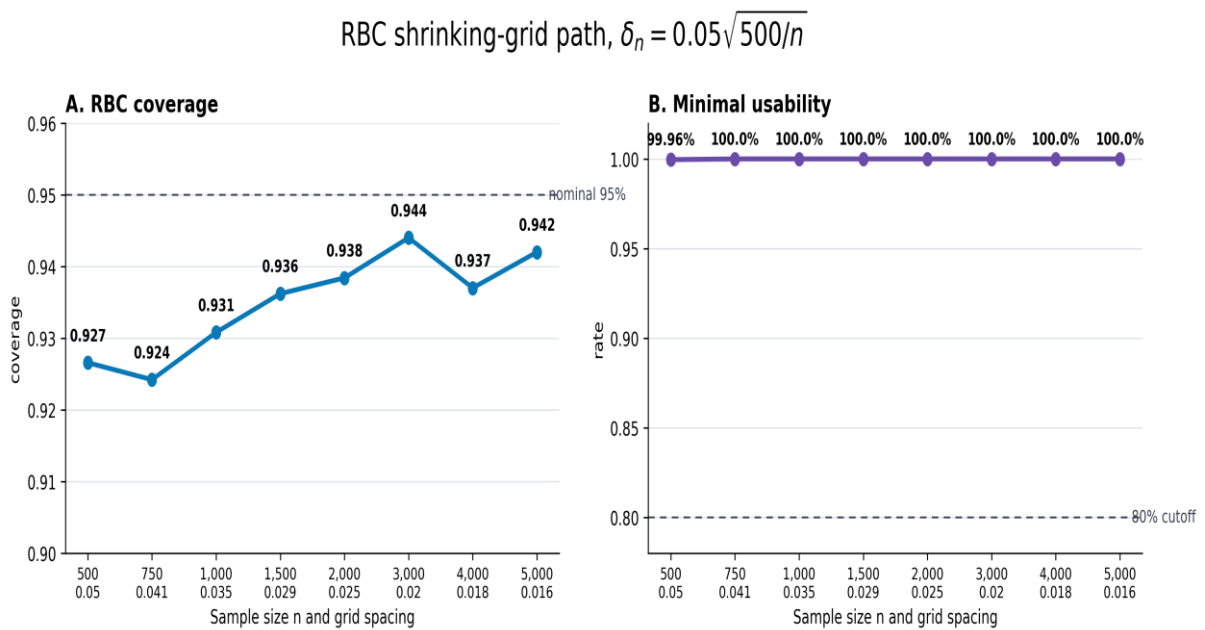


Figure 4.10. RBC performance along the shrinking-grid path

Figure 4.10 first shows that discreteness by itself does not immediately invalidate RBC inference. Along the shrinking-grid path, RBC coverage remains between 92.42% and 94.40%, increasing from 92.66% at $n = 500$ to 94.20% at $n = 5000$. Over the same range, the average interval length falls from 0.339 to 0.124, while the minimal-usability rate rises from 99.96% and then remains at 100%. When grid spacing shrinks with the sample size, the shorter intervals therefore reflect a normal gain in precision.

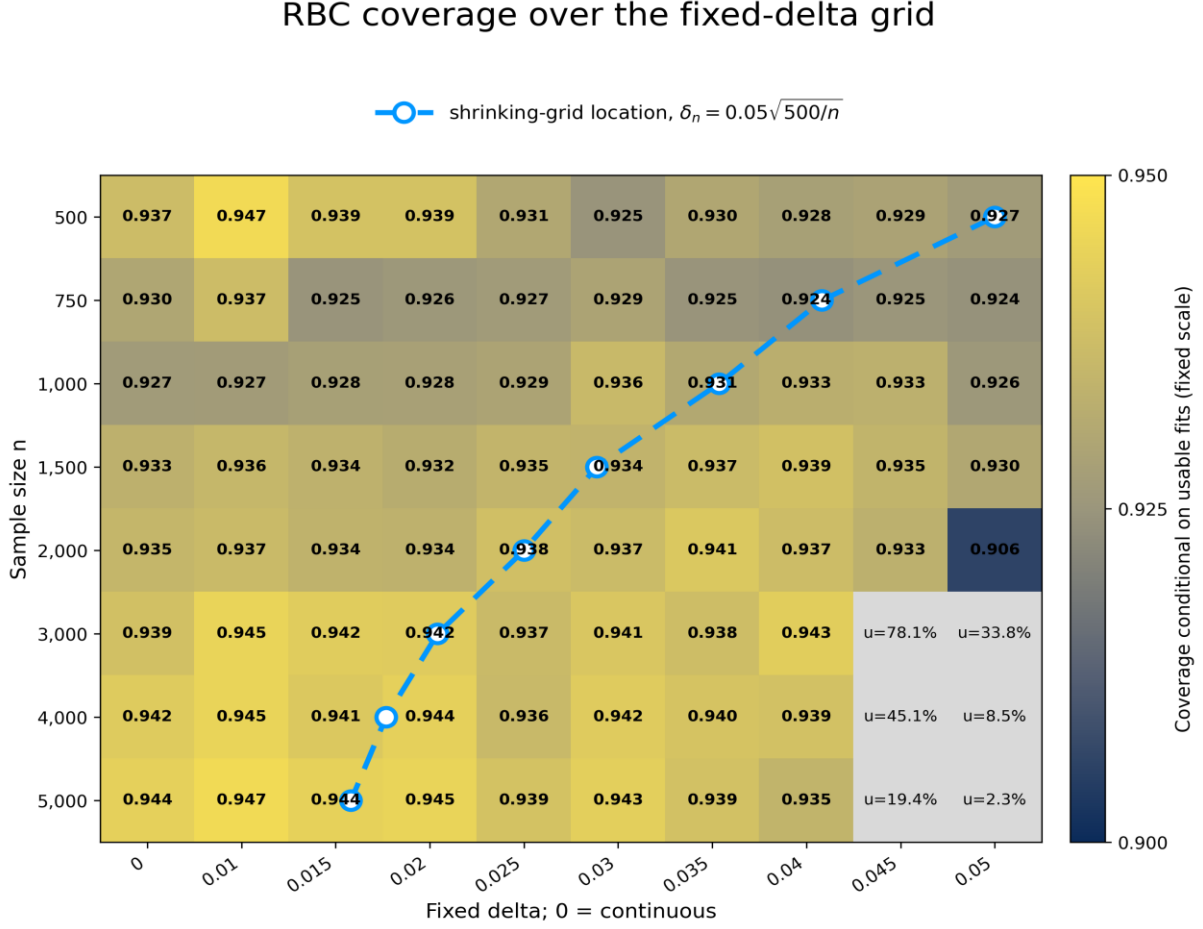


Figure 4.11. RBC coverage and aggregate usability over the fixed-delta grid

The fixed-support path, by contrast, produces an apparently counterintuitive pattern. For $\delta = 0.05$, the average interval length also falls from 0.339 and reaches 0.128 among minimally usable repetitions at $n = 5000$. Yet minimal usability declines from 99.96% to 2.28%, conditional coverage falls from 92.66% to 32.46%, and RBC bias rises from 0.023 to 0.076. Shorter intervals on a fixed grid therefore do not imply more precise inference.

The reason is that, under fixed support, additional observations accumulate at existing mass points without creating new values of the running variable. At the same time, the MSE-optimal bandwidth generally shrinks as n increases, causing h/δ and b/δ to decline. Thus, even as the number of observations within the bandwidth grows, the number of distinct support points it contains may fall. The operability of local polynomial estimation is therefore determined not only by the number of observations but also by whether sufficiently rich distinct support exists near the cutoff.

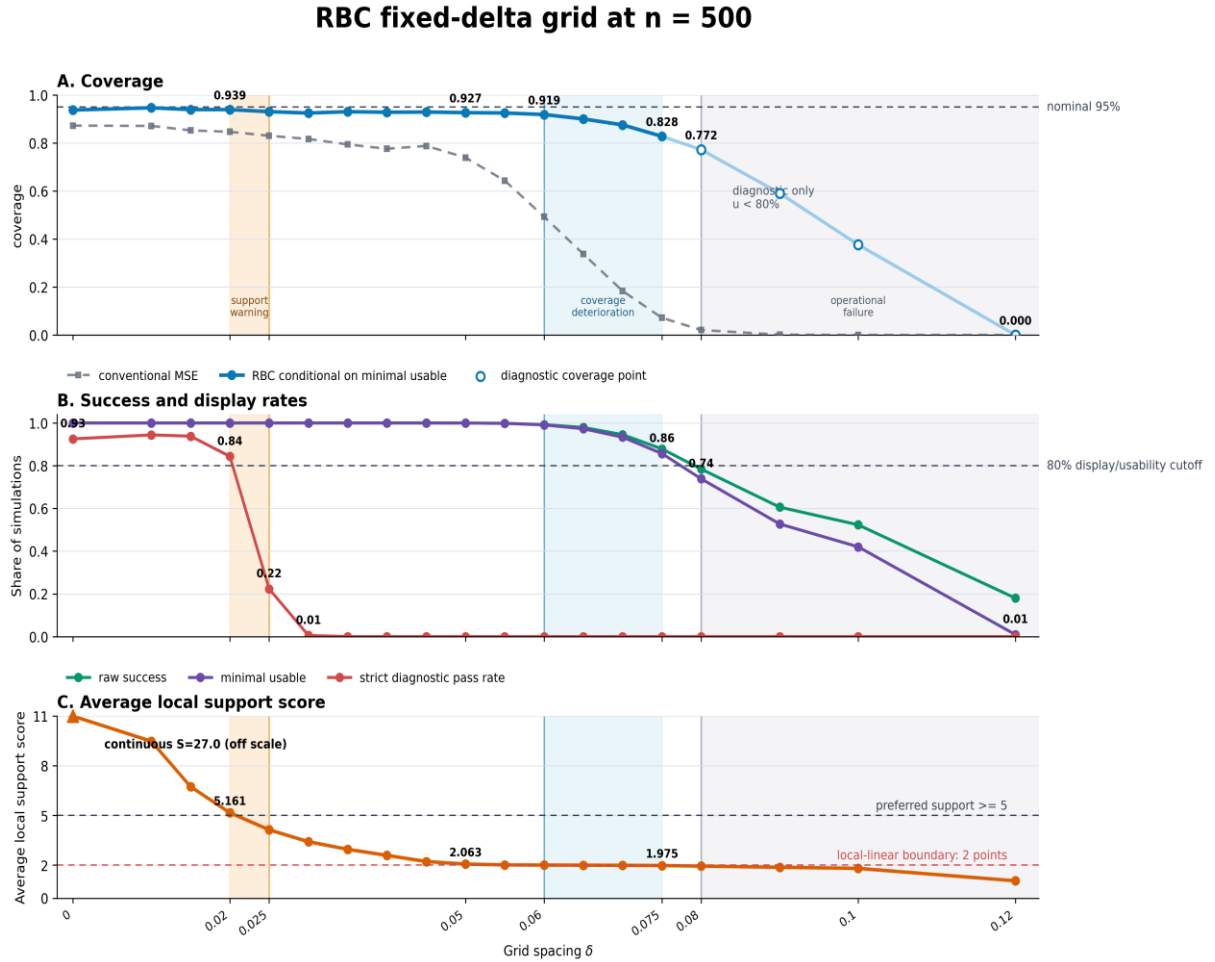


Figure 4.12. Discrete RBC failure frontier at $n = 500$

The frontier extension sharpens this result. Figure 4.12 shows that failure does not occur at a single threshold but unfolds through three regions: a support warning, coverage deterioration, and operational failure. At $\delta \approx 0.02$ -0.025, the preferred-support diagnostic has already deteriorated markedly, but minimal usability remains close to 100% and RBC coverage remains near 93%; this range is therefore an early-warning region rather than a failure region. Beginning at approximately $\delta = 0.06$, coverage and usability decline together. By $\delta = 0.08$, minimal usability falls below 80%, placing the design in the operational-failure region. The main mechanism is that residual bias grows rapidly as the grid becomes coarser, while the confidence interval does not widen correspondingly, causing noncoverage to become increasingly concentrated in one direction. Under coarse discrete support, RBC can no longer estimate and correct local curvature reliably.

Across the frontier, RBC nevertheless generally fails later than the Conventional CI. At $\delta = 0.05$, their coverage rates are 92.66% and 73.93%, respectively. At $\delta = 0.075$, RBC coverage remains 82.78%, whereas Conventional coverage has fallen to 7.16%. RBC retains this relative advantage because it can correct part of the smoothing bias. The advantage does not mean, however, that RBC can solve the problem of coarse discrete support. Once only a few distinct support points remain within the bandwidth, the local regression function itself cannot be estimated stably. Failure under fixed coarse support is therefore first a local-support and estimation problem, which subsequently deprives inference of its conventional interpretation.

Table 4.4. RBC Performance by Local Support Score

Local support score	RBC coverage	Minimal-usability rate	Preferred-support diagnostic pass rate
13+	93.59%	100.00%	92.76%
8–12	93.90%	100.00%	96.88%
5–7	93.59%	100.00%	94.71%
3–4	91.90%	100.00%	0.00%
0–2	85.20%	90.66%	0.00%

To relate RBC performance directly to the local support actually available, Table 4.4 pools all repetitions across the δ scenarios in the $n = 500$ failure-frontier experiment and classifies them according to the local support score S_r : 0–2, 3–4, 5–7, 8–12, with 13+ as the highest category.

When $S \geq 5$, RBC coverage remains approximately 93.6%-93.9%. When $S = 3-4$, coverage is still 91.90%, indicating that fewer than five support points should be treated as a warning rather than an immediate failure threshold. Once S falls to 0-2, however, coverage drops markedly to 85.20%, and operability problems become concentrated in this group.

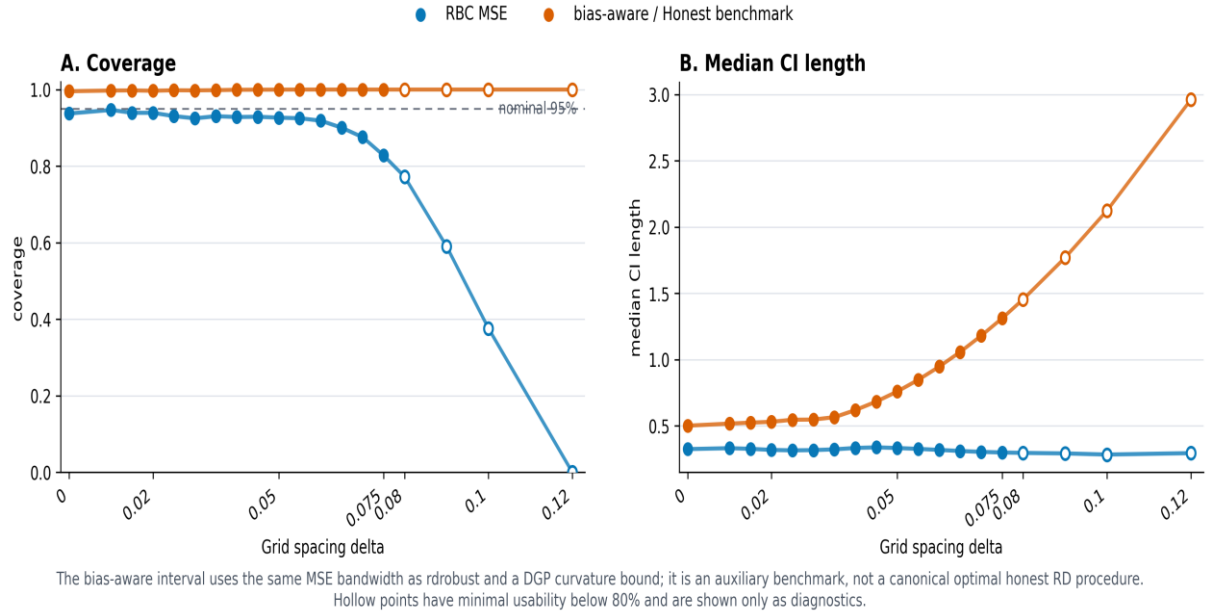


Figure 4.13. Bias-aware curvature-bound interval under coarse discrete support

Figure 4.13 reports the supplementary bias-aware curvature-bound interval. It preserves near-complete conditional coverage among minimally usable repetitions, but only by widening the interval substantially: the median length increases from 0.501 in the continuous design to 1.454 at $\delta=0.08$ and 2.961 at $\delta=0.12$. This illustrates the main logic of bias-aware inference: explicit allowance for worst-case curvature bias can protect coverage, but at the cost of wider intervals. Since minimal usability at $\delta=0.12$ falls to only 0.96%, however, this result should be read as a conditional sensitivity benchmark rather than as a solution to the coarse-support problem.

Overall, RBC can handle finely discrete running variables or grids that become denser with sample size, but it cannot overcome fixed coarse support when too few distinct support points remain near the cutoff. Under such designs, interval length may shrink with sample size even as residual bias and estimation failure risk increase. Empirical applications should therefore report not only observations within the bandwidth, but also distinct support points on both sides within h and b , together with usability diagnostics. Once the design enters a deterioration or failure region, bias-aware sensitivity analysis or, where appropriate, a local-randomization framework may be more suitable.

5. Discussion: Scope and Limits of RBC

Table 5.1. Summary of Coverage across Simulation Designs

Design	Conventional Coverage	RBC Coverage	Main Implication
Baseline	85.24%	93.00%	RBC substantially reduces undercoverage caused by smoothing bias.
Heteroskedasticity	84.5%–85.5%	92.2%–93.3%	Robust Studentization remains effective under different variance structures.
Non-normal errors	85.2%–87.4%	93.0%–95.9%	Error normality is not essential for RBC performance in the designs considered.
Shrinking-support path	—	92.42%–94.40%	RBC remains operational when the support becomes denser as sample size increases.
Fixed-support path	—	92.66%→32.46%	A larger sample cannot compensate for an insufficient number of distinct support points.
Failure frontier ($\delta=0.05\rightarrow0.075$)	73.93%→7.16%	92.66%→82.78%	RBC delays coverage deterioration, but cannot prevent failure as the discrete grid becomes increasingly coarse.

The results reveal two qualitatively different forms of pressure. Heteroskedasticity and non-normal errors alter the variance structure or the error distribution but do not disrupt local support or smoothness of the conditional mean near the cutoff. The problem therefore remains an inferential one within the local-polynomial framework, and RBC can continue to improve conventional inference through bias correction and robust Studentization.

A discrete running variable reveals a more fundamental boundary of RBC. When the grid becomes denser as the sample size increases, local support improves and RBC coverage remains close to the Baseline. Under fixed coarse support, by contrast, additional observations accumulate at existing mass points without increasing the number of distinct support points. As the data-driven bandwidth shrinks, usable local support may decline further. The issue is then no longer only one of confidence-interval construction; it is first a local-support and estimation problem. RBC can correct smoothing bias that can be estimated, but it cannot create local information that is absent from the data.

RBC-MSE can therefore serve as the primary inferential method under continuous or sufficiently fine support, and heteroskedasticity or non-normality alone generally does not require abandoning RBC. With a discrete running variable, researchers should additionally report the numbers of distinct support points on each side of the cutoff within both the main bandwidth and the bias-correction bandwidth and should inspect usability diagnostics. When local support is clearly insufficient, bias-aware inference can be used as a sensitivity analysis under credible curvature restrictions. If treatment assignment near the cutoff can reasonably be regarded as approximately random, a local-randomization framework may instead be appropriate.

The support-warning range, failure frontier, and local-support-score thresholds identified here depend on the particular DGP, sample size, bandwidth choice, and discretization path and should not be interpreted as universal critical values for all RD designs. Moreover, when operability declines, the reported coverage is conditional on usable repetitions. The results are therefore intended to reveal failure mechanisms rather than to provide universally applicable numerical boundaries.

6. Conclusion

Starting from CCT (2014), this paper uses a unified paired Monte Carlo design to examine how far the finite-sample advantages of RBC inference extend. The central question is not only whether RBC reduces the undercoverage of conventional inference, but also which conditions sustain that improvement and in which cases it ceases to have a conventional interpretation.

The simulation results provide a consistent answer. When heteroskedasticity and non-normal errors change only the sampling environment while local support near the cutoff remains adequate, the coverage advantage of RBC is broadly stable. When fixed coarse support causes the number of distinct support points to decline persistently, RBC moves from stable improvement toward eventual failure. The true boundary of RBC therefore lies not in whether the error distribution is ideal, but in whether local polynomial estimation itself retains an adequate informational basis.

The finite-sample performance of RBC can consequently be summarized as conditionally robust. RBC is an effective method for addressing smoothing-bias-driven undercoverage within the continuity-based RD framework, but its reliability depends on the local regression function remaining stably estimable.

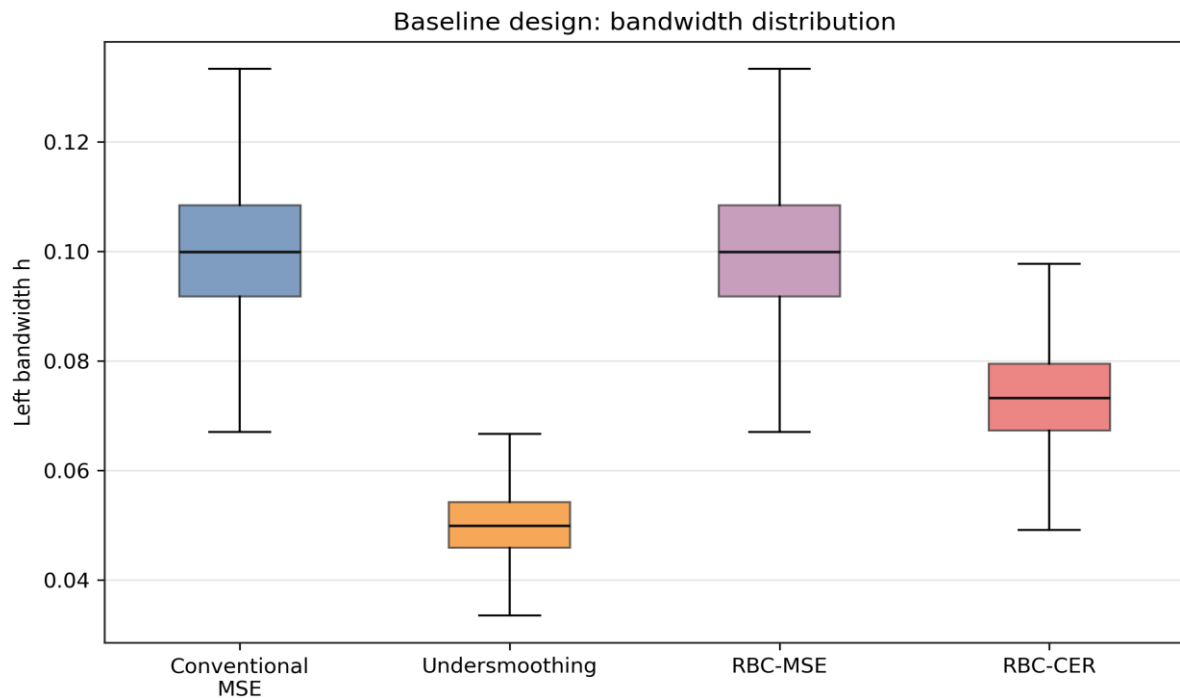
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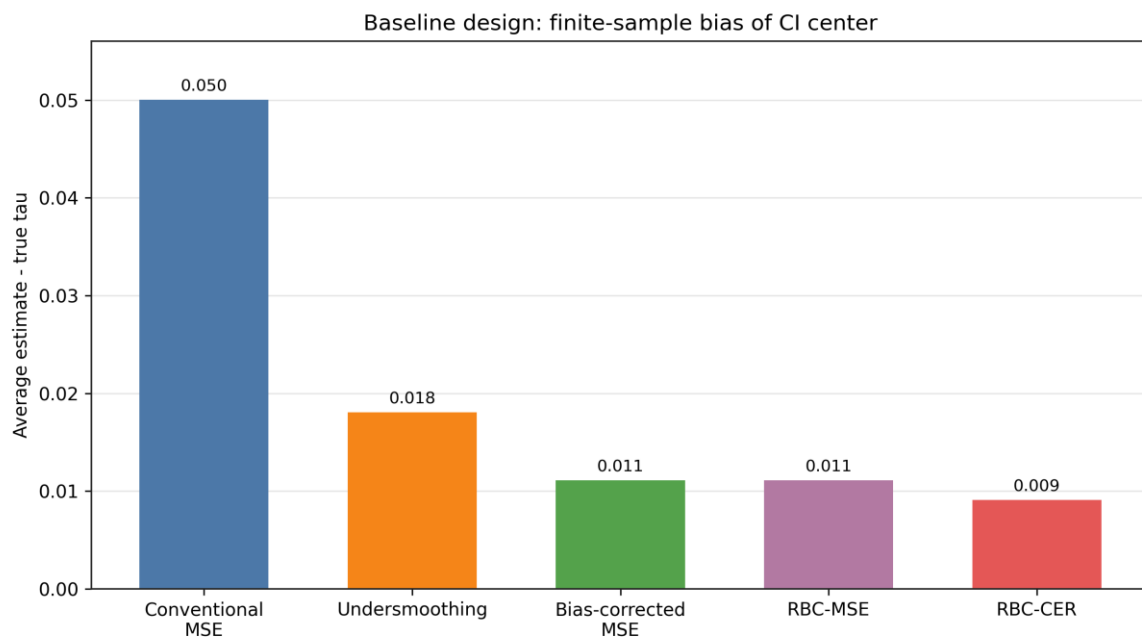
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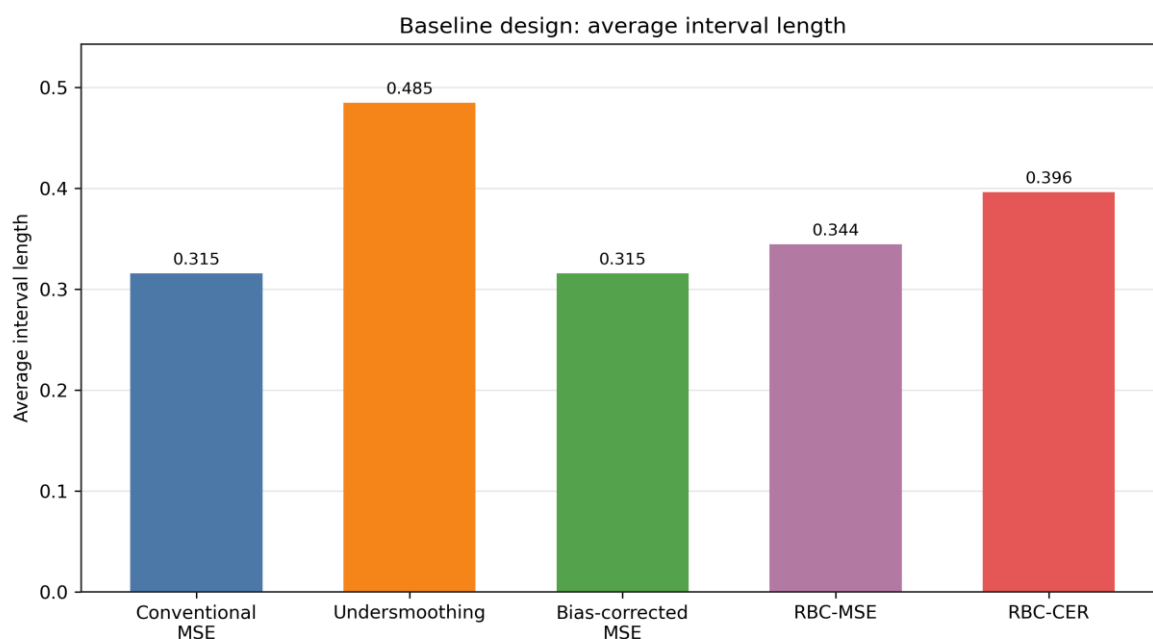
8. Appendix: Supplementary Figures



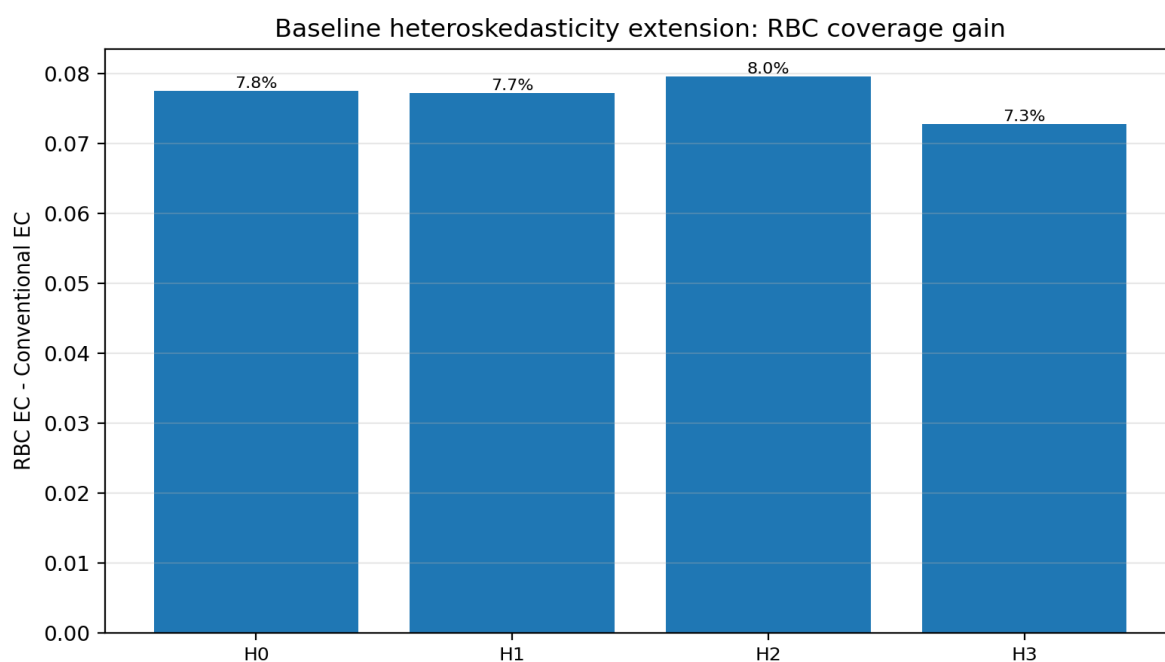
Supplementary Figure S.1. Baseline bandwidth distribution



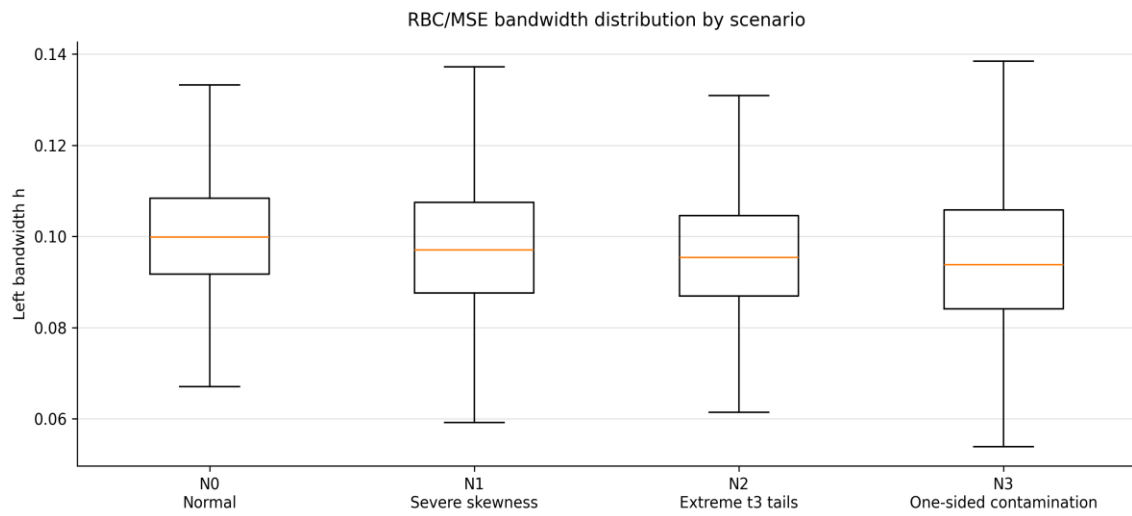
Supplementary Figure S.2. Baseline bias by method



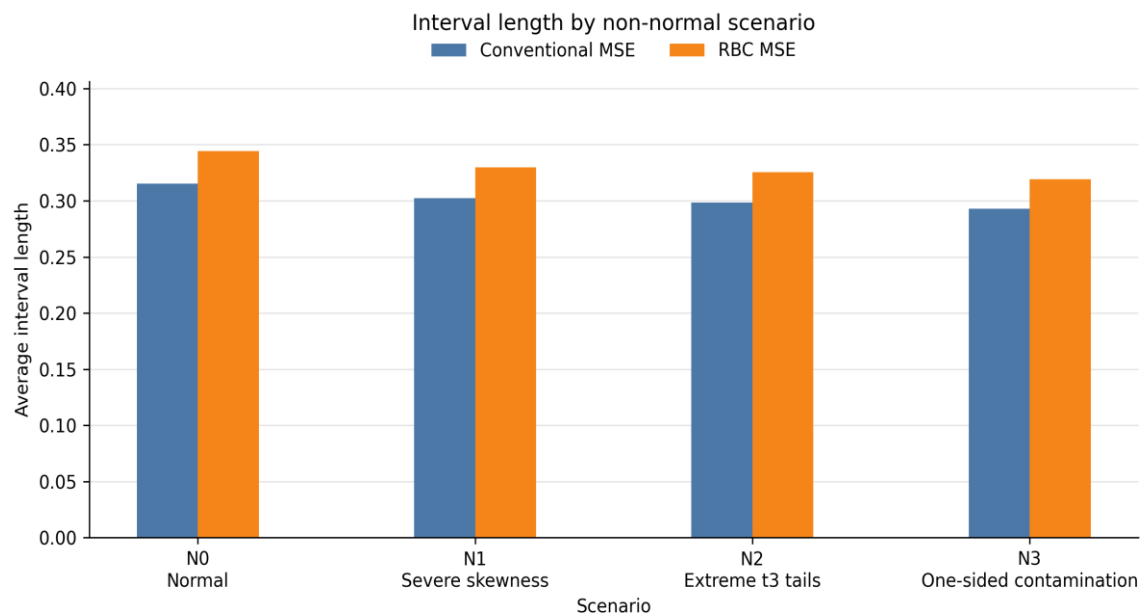
Supplementary Figure S.3. Baseline interval length by method



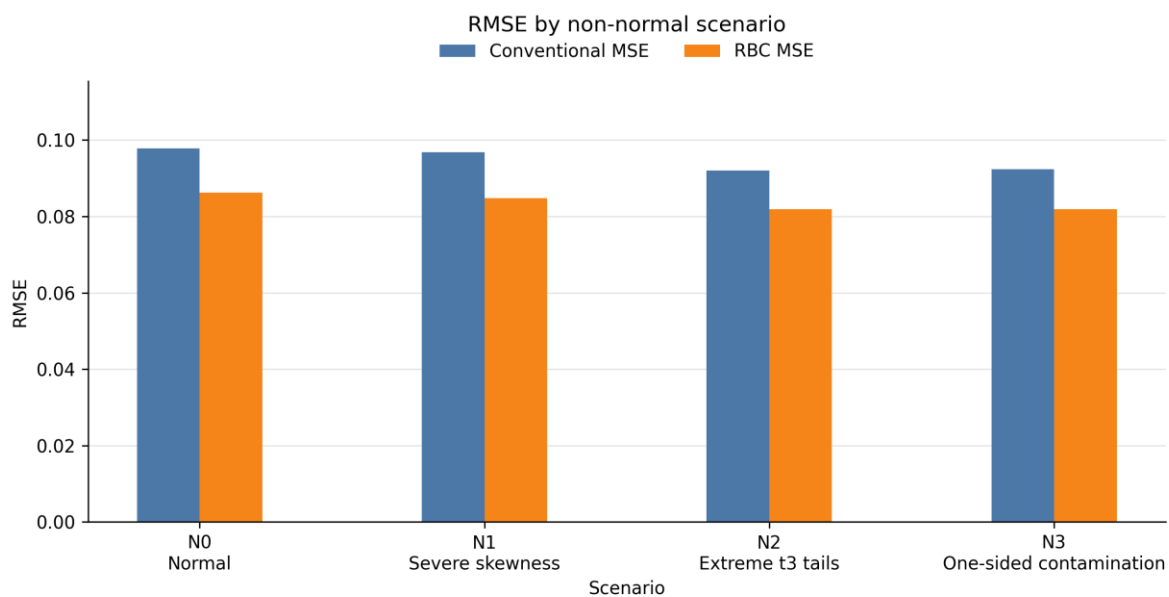
Supplementary Figure S.4. RBC coverage gain under heteroskedasticity



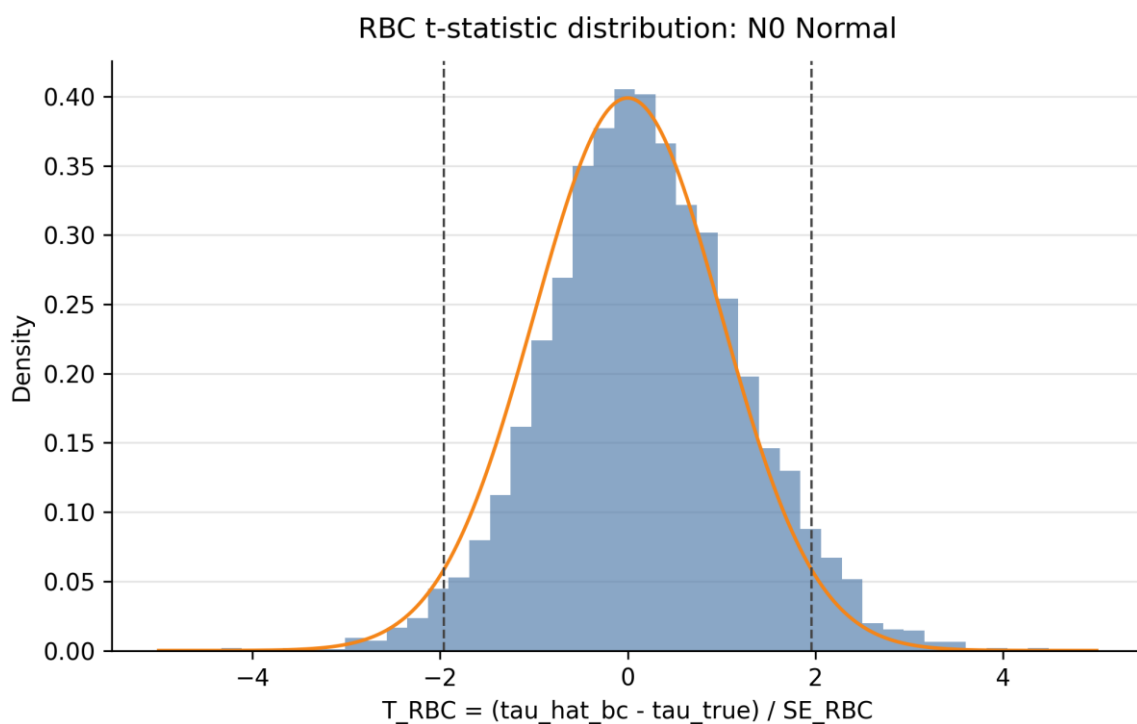
Supplementary Figure S.5. Distribution of selected left bandwidths under non-normal errors



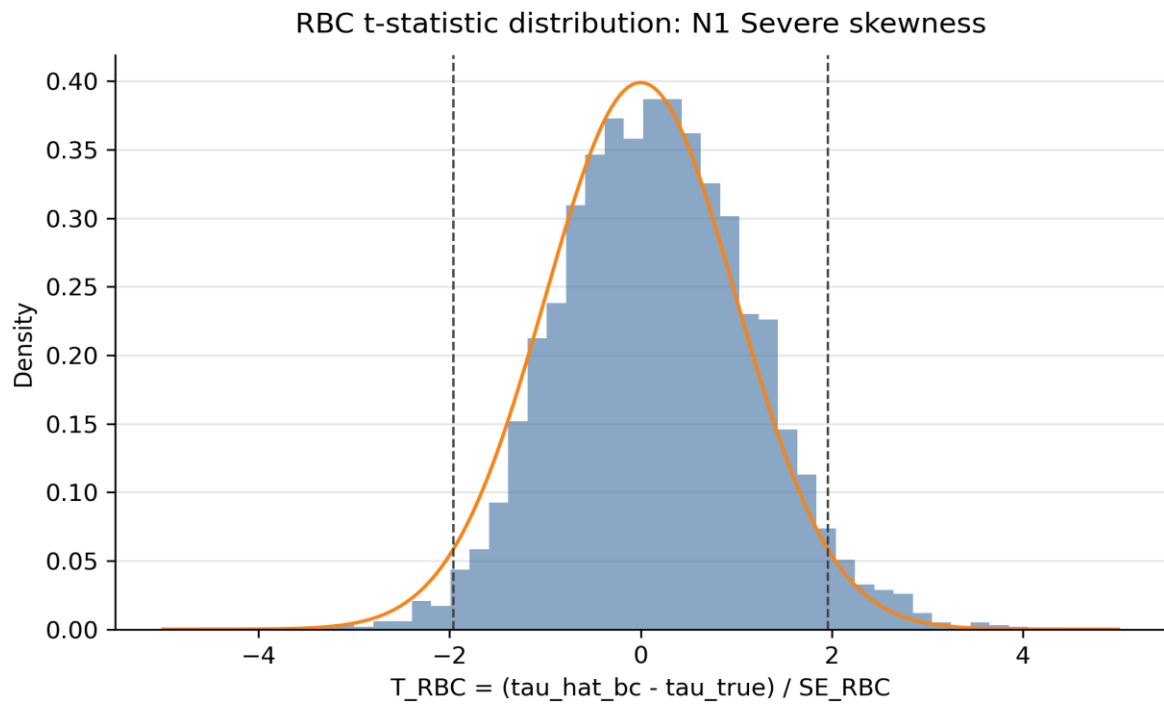
Supplementary Figure S.6. Interval length by non-normal scenario and method



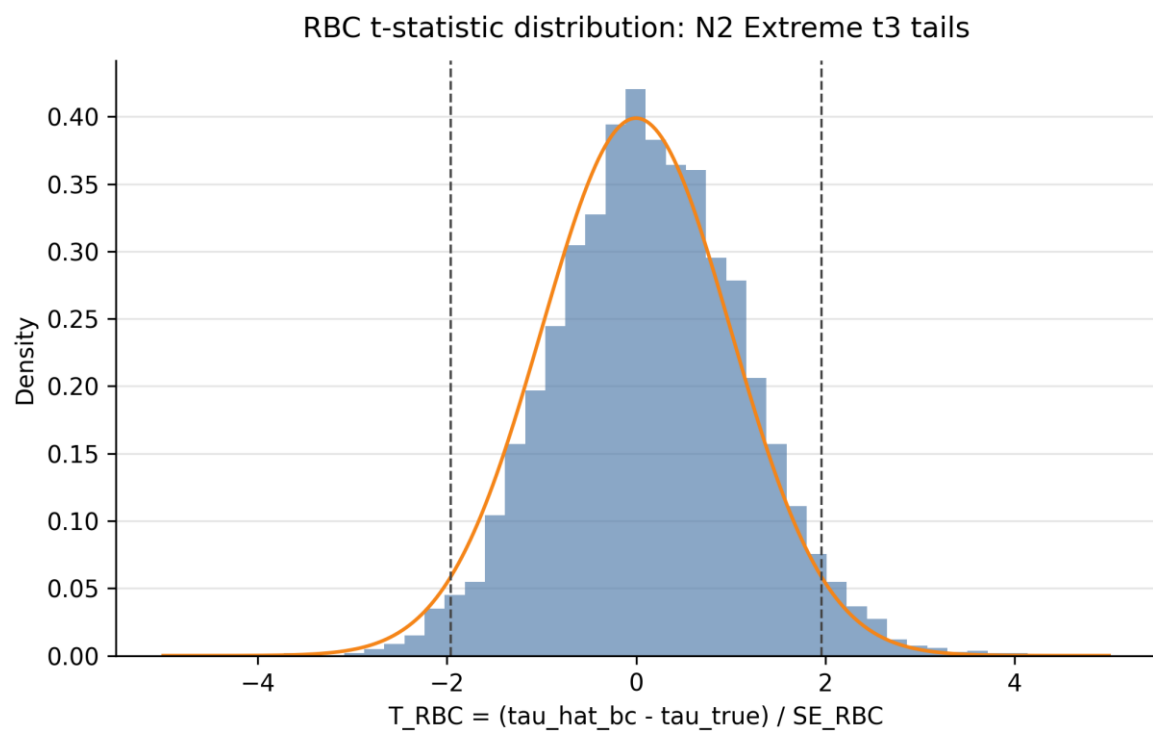
Supplementary Figure S.7. RMSE by non-normal scenario and method



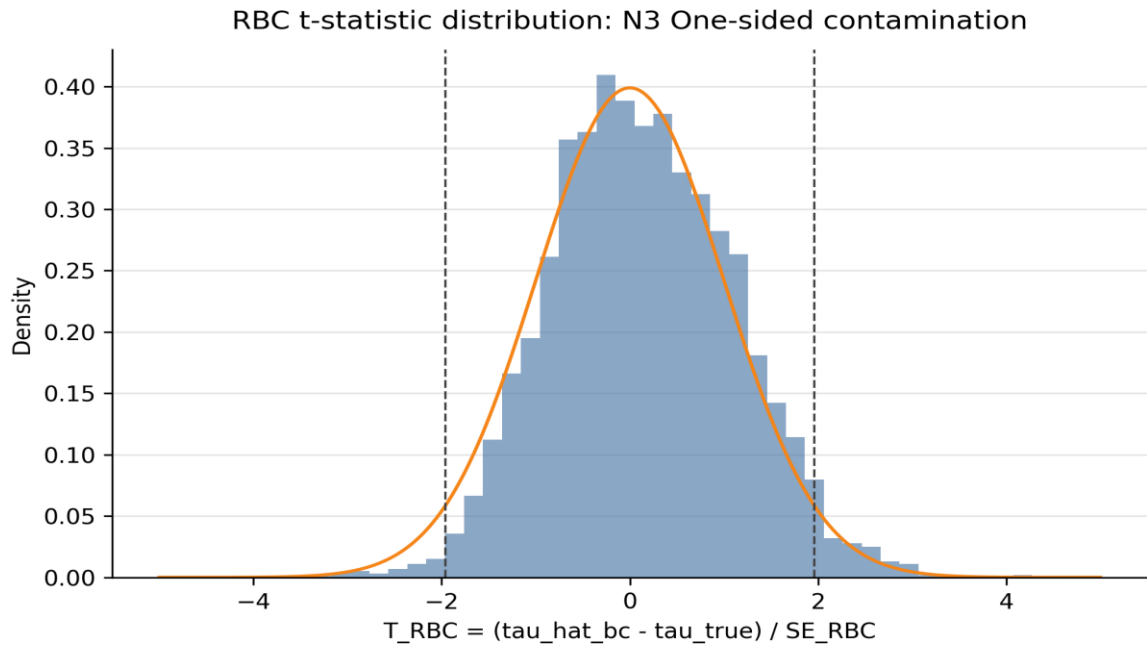
Supplementary Figure S.8. RBC t-statistic distribution under N0 normal errors



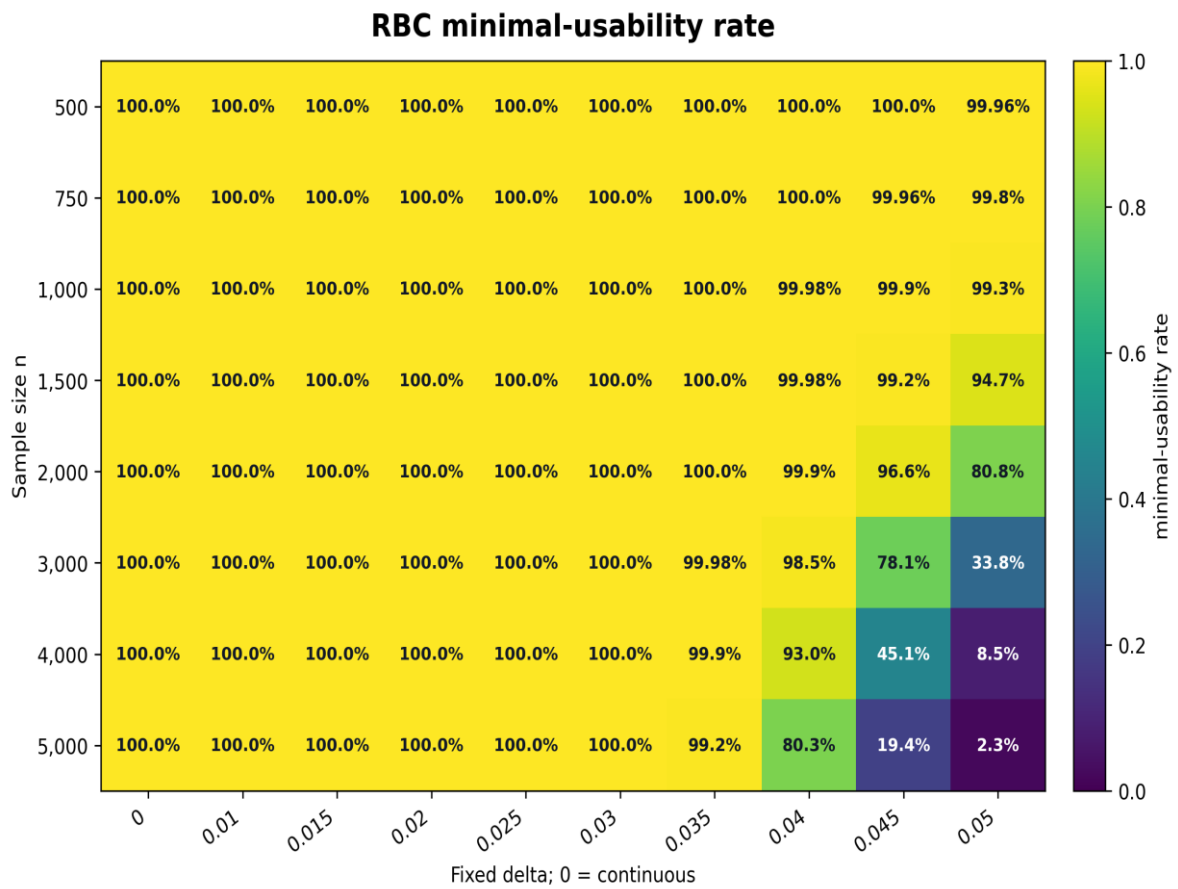
Supplementary Figure S.9. RBC t-statistic distribution under N1 severe skewness



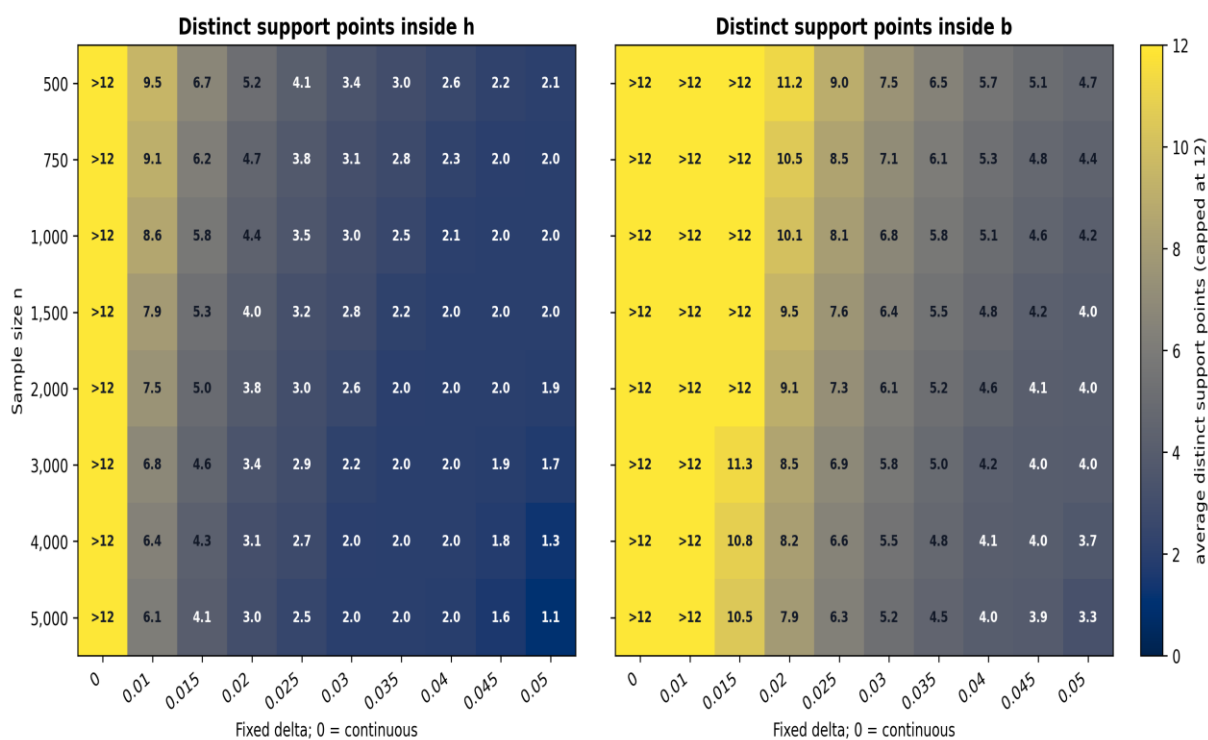
Supplementary Figure S.10. RBC t-statistic distribution under N2 extreme t3 tails



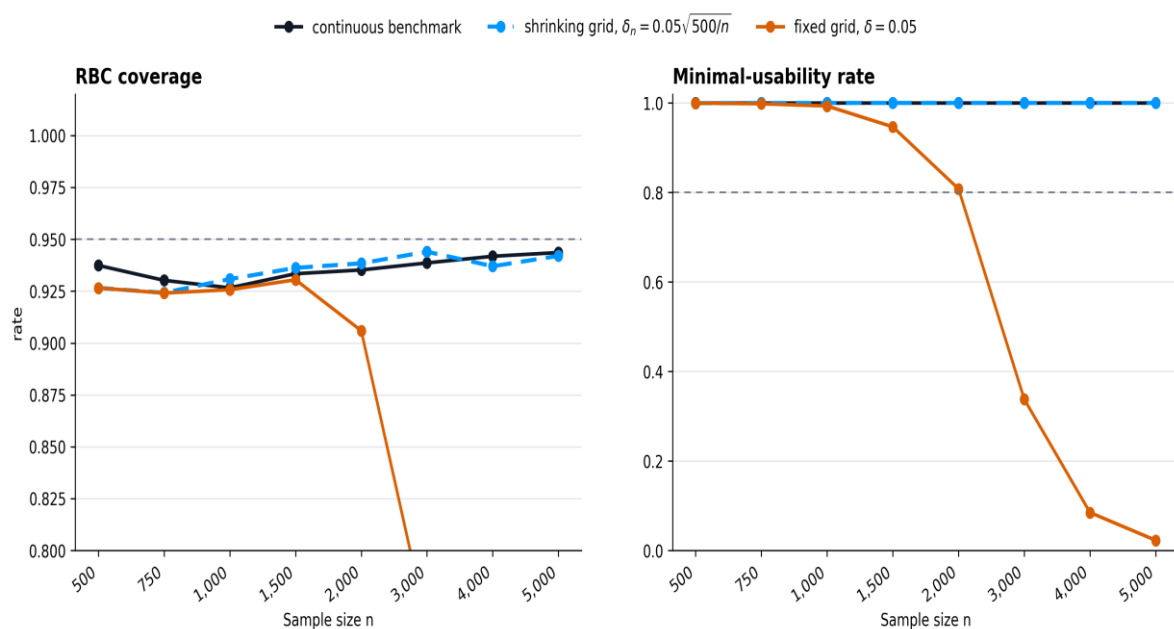
Supplementary Figure S.11. RBC t-statistic distribution under N3 one-sided contamination



Supplementary Figure S.12. RBC minimal-usability heatmap



Supplementary Figure S.13. Local support heatmaps



Supplementary Figure S.14. Fixed-grid performance compared with continuous and shrinking-grid benchmarks

Statement for the Use of Generative AI

During the preparation of this seminar paper, I used OpenAI Codex only for code-related assistance in the simulation implementation. I have reviewed and edited the code and related content as needed and take full responsibility for the content.

Name: Ziyuan Li

Date: 21/06/2026

Place: Munich